RESEARCH-ARTICLE

The Collaborative Sensemaking Play of Jubensha Games: A Deconstruction, Taxonomy, and Analysis

SHANO LIANG, Worcester Polytechnic Institute, Worcester, MA, United States

MAX CHEN, Worcester Polytechnic Institute, Worcester, MA, United States

PHOEBE O. TOUPS DUGAS, Monash University, Melbourne, VIC, Australia

GILLIAN SMITH, Worcester Polytechnic Institute, Worcester, MA, United States

ROSE BOHRER, Worcester Polytechnic Institute, Worcester, MA, United States

Published: 31 March 2025
Online AM: 28 February 2025
Accepted: 07 February 2025
Revised: 17 December 2024
Received: 04 July 2024

The Collaborative Sensemaking Play of Jubensha Games: A Deconstruction, Taxonomy, and Analysis

SHANO LIANG, Interactive Media and Game Development, Worcester Polytechnic Institute, Worcester, United States
MAX CHEN, Interactive Media and Game Development, Worcester Polytechnic Institute, Worcester, United States
PHOEBE O. TOUPS DUGAS, Human-Centred Computing / Faculty of Information Technology / Exertion Games Lab, Monash University, Melbourne, Australia
GILLIAN SMITH, Interactive Media and Game Development, Worcester Polytechnic Institute, Worcester, United States
ROSE BOHRER, Computer Science, Worcester Polytechnic Institute, Worcester, United States

Jubensha games, popular in China, combine storytelling, social deduction, and script-centered group play, sparking widespread interest among gamers and researchers worldwide. However, enthusiasts and researchers have struggled to accurately describe *Jubensha*, often defaulting to comparisons with genres like murder mystery and live-action role-playing games. This reliance on comparisons hinders efforts to generalize *Jubensha* or to deconstruct and adapt its unique design components, dynamics of player interaction, and playing experience into other games. This research provides a taxonomy and analysis of *Jubensha* games, based on a thematic analysis of over 80 *Jubensha* games accessed through mobile applications and physical copies. The analysis combines the authors' positionalities as native Chinese and English speakers and lenses from close reading of the games, discourse analysis, and distributed cognition. We provide summative case studies to exemplify our taxonomy and discuss design implications for *Jubensha* games for future projects. Our work provides a descriptive tool and vocabulary for researchers and designers to facilitate communication and theorize the evolving *Jubensha* gaming phenomenon highlighting how gameplay centers collaborative sensemaking. In addition, we argue that the design of game narratives in *Jubensha* games, including structures in scripts and the evolving performance of players, can be generalized and transferred to the design of a wider range of analog and video games.

CCS Concepts: • **Human-centered computing → Qualitative studies in HCI**; **HCI theory, concepts and models;**

Additional Key Words and Phrases: Jubensha, murder mystery games, taxonomy, game analysis, game design, cross-cultural context

Authors' Contact Information: Shano Liang, Interactive Media and Game Development, Worcester Polytechnic Institute, 100 Institute Rd, Worcester, Massachusetts, United States; e-mail: sliang1@wpi.edu; Max Chen, Interactive Media and Game Development, Worcester Polytechnic Institute, 100 Institute Rd, Worcester, Massachusetts, United States; e-mail: mjchen@wpi.edu; Phoebe O. Toups Dugas, Human-Centred Computing / Faculty of Information Technology / Exertion Games Lab, Monash University, Faculty of Information Technology, 25 Exhibition Walk, Clayton, Victoria, Australia; e-mail: phoebe.toups.dugas@acm.org; Gillian Smith, Interactive Media and Game Development, Worcester Polytechnic Institute, 100 Institute Rd, Worcester, Massachusetts, United States; e-mail: gmsmith@wpi.edu; Rose Bohrer, Computer Science, Worcester Polytechnic Institute, 100 Institute Rd, Worcester, Massachusetts, United States; e-mail: rbohrer@wpi.edu.

ACM 2832-5516/2025/03-ART6
https://doi.org/10.1145/3721121

ACM Reference Format:
Shano Liang, Max Chen, Phoebe O. Toups Dugas, Gillian Smith, and Rose Bohrer. 2025. The Collaborative Sensemaking Play of Jubensha Games: A Deconstruction, Taxonomy, and Analysis. *ACM Games* 3, 1, Article 6 (March 2025), 34 pages. https://doi.org/10.1145/3721121

1 Introduction

剧本杀 (*Jubensha*) is an emergent category of script-centered group gaming phenomena that centers collaboratively coming to understand an unfolding series of fictional events—an experience of distributed cognition and sensemaking—unlike games that prioritize winning and losing states. Playing a *Jubensha* game typically involves four to eight players co-located in board game cafes or a combination of co-located and/or remote players on mobile devices (with voice communication when remote). Players progress with a series of cold script readings and group interactions to uncover and experience the story. Most *Jubensha* games also include mechanics that encourage competition between players or between players and elements of the environment.

Jubensha has gained tremendous popularity in mainland China since the late 2010s [95, 105, 106], with thousands of *Jubensha* stores, board game cafes, and digital *Jubensha* game platforms opening up [34]. A 2021 report on physical *Jubensha* consumer insight by Meituan [44] forecast sales of RMB 15.42 billion (~USD 2.2 billion) to 9.41 million consumers.

There is an emerging interest in research into *Jubensha* games, as the essential design components and player experience that can further inform game design. We provide insights into the *Jubensha* gaming phenomena for global audiences and explore its characteristics, taxonomy, and design implications.

1.1 Contribution

We provide designers and game researchers with a granular framework for describing, experimenting with, and applying *Jubensha* games or their components in diverse contexts. Our research aims to break down the key features and combination of elements in *Jubensha* games, with particular emphasis on their script-centered play. We contribute a ludography of *Jubensha* games—83 games—that we played and examined.

While Chinese scholarship has explored potential applications of *Jubensha* games in non-game contexts (e.g., References [94, 95, 102, 105, 106, 109, 111–113]) and in global venues (e.g., References [55, 58]), as well as examining player experience (e.g., Reference [99]), the research community has no clear agreement and consensus on the definition of *Jubensha*. Scholars and journalists often oversimplify *Jubensha* by focusing either on the detective themes (e.g., concluding it is a variant of murder mystery games) or live-action elements (e.g., a subset of live-action role-playing (larp) games). These mis-characterizations restrict the exploration of *Jubensha*'s unique characteristics and player dynamics inherent in its script-centered group interactions.

Existing game taxonomies focus heavily on player interaction and choices [1, 2, 5, 30, 65, 74, 92] and center a definition of "game" in which designed choices (mechanics) allow players to make decisions that drive toward an uneven outcome [45, 76]. We develop two layers of meaning of our taxonomy: locating of a set of *Jubensha* games in existing taxonomies by deconstructing *Jubensha* games into their essential features and a new taxonomy within the set of *Jubensha* games.

When we consider *Jubensha* among prior taxonomies, *Jubensha* games are primarily played by engaging in social and group sensemaking of stories in a distributed cognition environment. In *Jubensha* play, *information distribution* [73, 90]—wherein players have access to alternative pieces of a complete information puzzle—is a key element. Figure 1 shows the physical elements of *Plural*

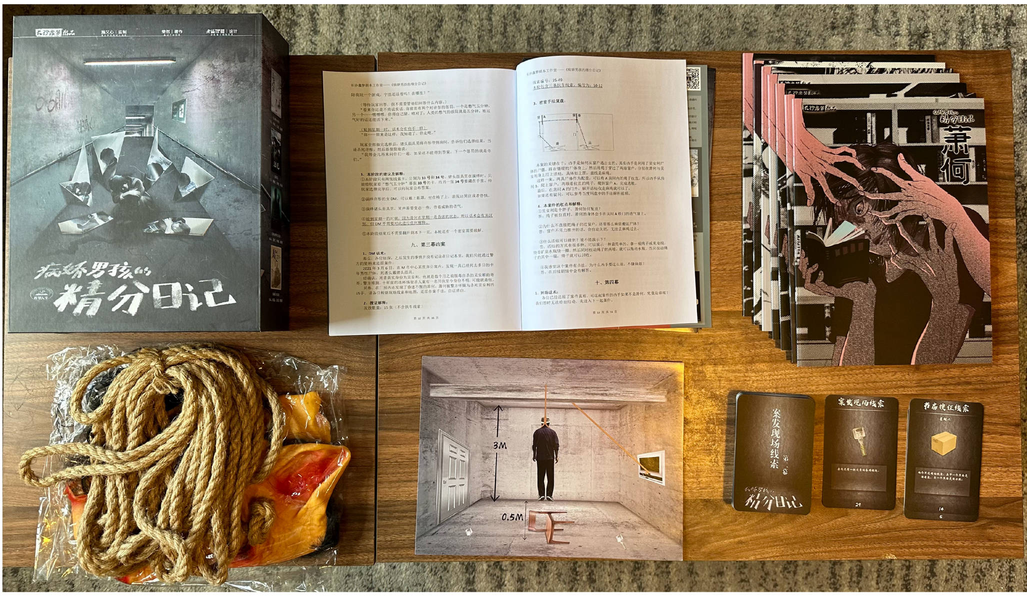

Fig. 1. Photo of the physical version of the *Jubensha* game, The Plural Diary of the Yandere Boy [G37], depicts the game's INTERFACE with integration through the MECHANICS (read left-to-right, top-to-bottom). **Upper Left:** The game packaging. **Upper Middle:** The game host's manual. **Upper Right:** Player scripts. **Lower Left:** Props for the TABLEREADING of the game. **Lower Middle:** Props for PLAYER DISCUSSION and INVESTIGATION. Included here are pictures of the crime scene and related components. **Lower Right:** Information cards and item cards for INVESTIGATION of the game. Photographed by the authors.

Diary of the Yandere[1] *Boy* [G37], which are used to distribute information among players. Game characters, objects, events, and worlds are predefined and provided to players in their scripts, which each player must interpret and selectively reveal to others. The *core mechanics*—the main choices that players make [76]—center communication [89] and carefully engaging in distributed cognition [38, 40, 73]—accessing and deciphering information and changing its representation to share with others.

For the taxonomy of different types of *Jubensha* games, the Chinese community of *Jubensha* designers, players, and press can provide some insights through their discussions. These generally address setting aesthetics (e.g., *Japanese-styled*, *European-styled*), era background (e.g., *Modern*, *Medieval*) and other setting elements (e.g., *Magic*, *Time-travel*). Such discussions do not directly contribute to constructing the gameplay and narrative structure of *Jubensha*, but offer an initial framing used by many people. Some *Jubensha* stores and online apps categorize *Jubensha* games based on players' feedback on plots and/or core mechanics, but these categories are used interchangeably, do not always have agreement within the community, and do not represent a consistent taxonomy. The lack of precision in describing and classifying *Jubensha* makes it challenging for researchers to make progress in understanding, translating, and exploring new approaches in *Jubensha* as well as other storytelling-heavy games.

[1]The phrase "*Yandere*" refers to someone who first exhibits affection for someone they are attracted to, but then develops intense and psychologically harmful tendencies. The genesis of this phrase may be traced back to Japanese anime, which combines the Japanese terms "*yanderu*" (病んでる) and "*deredere*" (でれでれ). For detailed explanations, see Kroon's work [51, p. 760].

1.2 Research Questions

Developing an understanding of the core mechanics and essential features of *Jubensha* games will support evaluating their implications for game design practice; their potential for use in other venues (e.g., References [95, 105, 107, 109]); and enable future researchers to conduct related projects, research, and designs. Driven by our curiosity and interest in *Jubensha*, our positionality of having deep knowledge of the scholarship and popular conversations surrounding it, and our motivation to support game designers and researchers in understanding this research space, we pose the **research question (RQ)**:

> **RQ1:** What are the key features of *Jubensha* games that enable their definition and analysis?

In engaging in the present research on deconstructing and analyzing *Jubensha* games, we realized the flexibility and usability of *Jubensha* games' framework allow future designers or researchers to use it for quick experience-oriented design prototyping. We aim to support researchers and designers in creating, using, and analyzing data from *Jubensha* for understanding and developing new forms of play. We thus identified an emergent second research question:

> **RQ2:** How can the key design features of *Jubensha* games contribute to other games' design?

1.3 Approach and Scope

The present research is exploratory—prior research has not sufficiently answered these questions. Further, the research topic is complex and nuanced, requiring an understanding of language and game design. To examine the gameplay and narrative of *Jubensha* games, we employ reflexive thematic analysis [18], a qualitative approach. While *Jubensha* was originally played physically at board game cafes and stores, digital mobile versions have become immensely popular. The present research mainly focuses on digital *Jubensha*, while the investigation also involves several physical *Jubensha* games for a thorough understanding. We choose to analyze games as our main source of data, because there is relatively little literature that addresses *Jubensha*'s definition and taxonomy. Our reflexive thematic analysis combines our positionality as native Chinese and English speakers and game designers, with the use of three lenses of close reading of games, discourse analysis, and distributed cognition.

We define *Jubensha* as a category of script-centered multiplayer games with the end goal of uncovering the unknown of the story and experiencing the narrative by group collaborative understanding and sensemaking in distributed cognition of the plot through information sharing, analyzing, deciphering, or implying from scripts. To reify our understanding of *Jubensha* game features and render it useful, we build up concrete design and analysis themes inductively from our analysis and deductively by incorporating the established ***Design, Dynamics, Experience*** **(*DDE*)** game design framework [93]. Moreover, we further contribute five brief summative case studies of popular *Jubensha* games under our themes.

This work contextualizes *Jubensha* in the intersectional realm of multiplayer games, role-playing games, murder mystery games, and interactive detective novels. It also contributes to the development of a shared vocabulary for *Jubensha* enthusiasts, offering a foundation for designing *Jubensha* games as well as evaluating player engagement and motivation.

1.4 Positionality Statement

We provide the positionality of the research team to support the reader in understanding the foci and perspectives we bring to the present work, as reflexive thematic analysis expressly calls the researchers' framing to bear [18]. Our framing is critical to the work. Researchers with different positionality and/or other corpus data items may identify different characteristics than we do.

The present research was undertaken by a team of five researchers—four in the USA and one in Australia. The two lead authors were born and raised in China; two team members are from the USA; the team member in Australia spent much of their life in the USA. This project relied heavily on language skills, primarily Mandarin Chinese but also Japanese. The two lead authors are native Mandarin speakers; three authors are native English speakers; a fifth author's Mandarin proficiency is estimated at HSK5-level[2] and their Japanese is certified at JLPTN4-level.[3]

All of the team members are well-played [4, 26]—we bring decades of experience playing video-, role-playing-, and board games. We also have limited experience using larp for design. We have been immersed in the discourse and culture surrounding games, their design, and their development. Further, all of the team members have experience in interaction design.

The two lead authors are well-played *Jubensha* enthusiasts. One of them has four years of experience in founding and managing a tabletop game company in China. They are deeply familiar with the games themselves, the culture surrounding them, and the current social media landscape in which these games are situated.

1.5 Global Context and Localization

The present research investigates games developed in China that are heavily text-based. All *Jubensha* games are written in simplified Chinese and situated within Chinese culture. Simple translation of the game names, contents, developers' names, and so on ranges from extremely difficult to literally impossible. Throughout the article, we ensure that the reader has context, since we can provide it—this includes translation, paraphrasing, and relevant cultural phenomena.

Games' information in Ludography is kept to their original simplified Chinese names to minimize information loss. We provide transliteration into Roman characters to enable English readers to follow more easily; where possible and useful to the reader, we provide translation and/or explanation.

In this article, the Chinese term "剧本杀" is transliterated into *Jubensha*, as a collective term for all games in this category.

1.6 Spoiler Warning

Many *Jubensha* games are designed and intended to be played once. That means once the *Jubensha* game contents (i.e., specific game rules and narratives) are revealed to players, they are not recommended to play that *Jubensha* game again. In Appendix A, we provide case studies on five *Jubensha* games that contain detailed descriptions of the games.

2 Background and Related Work

In this section, we start by providing background on existing literature about playing *Jubensha* in China. We contextualize *Jubensha* by comparing it with role-playing games, murder mystery games, and detective fiction, because they are the historical and practical dimensions of the study. Notably, while we discuss *Jubensha* in the framing as a specific Chinese gaming phenomenon with distinctive characteristics, we acknowledge that this study does not imply the absence of similar games beyond China. We then briefly address game design by introducing the the DDE Framework [93], which is central to the project and serves as the vector by which it may be taken up by designers and researchers. We finally introduce sensemaking in distributed cognition, because

[2]HSK stands for Hanyu Shuiping Kaoshi (version 2). HSK5 is the second-highest level of 6.
[3]JLPT stands for Japanese Language Proficiency Test, and N4 for "new level 4." This is the second-lowest level of 5.

they serve as our lens for analyzing the dynamics in *Jubensha* and a way to understand players players' interactions with *Jubensha* gameplay and narrative.

2.1 Jubensha Primer

Player communities assert that precursor games of *Jubensha* include *Death Wears White* [55, 62], *Mafia* [37, 60], and *Werewolf*-styled [27, 34] board games. These games involve social deduction and suspension of disbelief, captivating players with their intricate gameplay dynamics and the challenge of uncovering hidden identities. The burst in *Jubensha* began with the 2016 reality show *Who's the Murderer* on *Mango TV* [105]. In the show, celebrities role-played in a murder-mystery setting [34, 101].

Recently, journalists (e.g., Reference [67]) and scholars have shown interest in *Jubensha*. Chinese literature focuses on the applications of *Jubensha* gaming experiences in non-game contexts. These include, for instance, functions of *Jubensha* games in interactive film [105], theater performance [94], storytelling in tourism [102], addressing post-*COVID-19* collective solitude [112, 113], participatory education activities [95, 104, 107, 109], social or dating applications [111], promoting minority cultures [103], and developing group empathy [106]. Some relevant work addresses the trend of *Jubensha* industry [54], player categories and their situatedness in culture [99], and educational potentials of *Jubensha* [58]. However, these works have no clear agreement on the core mechanics and essential features of *Jubensha* games. Liu and Yang [54] address an industrial perspective on the evolution of *Jubensha* and storytelling styles of murder- and detective-focused *Jubensha* stories. While valuable to us to contextualize the history and storytelling of *Jubensha*, prior work is insufficient for understanding non-murder-involved *Jubensha* games.

Existing research has not yet formalized the taxonomy and definition of essential but distinctive features of the design and player interaction of *Jubensha*. Instead, scholarly discourse around *Jubensha* often mixes questions of uniqueness and derivation. By terming *Jubensha* a derivative of other gaming phenomenons due to their similarities (e.g., Western murder mystery games, larp), discourse not only offers a misleading understanding but also restricts potentials for exploring and analyzing the unique traits of *Jubensha* games. For the goal of this study and to contextualize *Jubensha*, we address *Jubensha* games' similarities and differences with role-playing games, murder mystery games, and detective fiction in the following subsections.

2.1.1 Role-Playing Games. ***Role-playing games*** **(RPGs)** are a heterogeneous set of games that share the distinguishing characteristic that a player steps into the role of a character in a story, enacting the character's choices [100]. Prominent *forms* of **RPG include tabletop (TTRPG)**, **live-action (larp)**, and RPGs that are fully played in software: single-player **computer RPGs (CRPG)** and **multiplayer online RPG (MORPG)** [100]. *Jubensha* games share certain features with these RPG forms: in-person *Jubensha* game playing often involves joint talk and paper inscriptions, resembling TTRPGs; in-store *Jubensha* games sometimes provide costumes to enhance immersion and materiality, echoing larp; and online *Jubensha* games have elements in common with MORPGs in terms of their digital environments and social interactions.

Notably, it is necessary to discuss larp here, because it is often (e.g., References [58, 110]) compared with *Jubensha*. Harviainen [35] defines larp as role-playing in which players physically presence and embody their characters within a fictional reality, and any disruption of that reality is considered a breach of the play. By synthesizing literature (e.g., References [63, 76, 81]), Harivianen's work [36] also highlights larp's general principles, such as efforts in maintaining the character separate and fictional, the temporary superimposition of fictional worlds on reality, and the pursuit of the "*360 degree aesthetic*" through the focus on immersion and materiality.

Despite these similarities, *Jubensha* games have traits distinct from these prominent forms of RPG. *Jubensha* is script-centered rather than character- and immersion-centered. These scripts are always fixed, leaving limited room for players to enact character-driven choices or deeply inhabit their roles—all elements of *Jubensha* serve the narrative structure of the prewritten scripts. For instance, in some *Jubensha* games (e.g., Games [G25, G61][4]), players may switch characters during the course of play, while in others (e.g., Games [G22, G30, G35, G40, G65]), characters might lack detailed portrayals or backgrounds, reduced to mere names or identifiers. Certain *Jubensha* games (e.g., Games [G17, G36–G38, G76]) even forgo characters entirely, encouraging players to review the story as external observers rather than active participants. In addition, *Jubensha* does not primarily focus on physical presence, immersion, and materiality.

2.1.2 Murder Mystery Games. Murder mystery games center players solving a mystery using clues and come in a variety of forms. In 1935, *Jury Box* [69], the first known published mystery dinner game including six cases, allowed players to act as jury members, examine the evidence against an accused person, and work to unravel the mystery to determine the defendant's guilt or innocence. *Cluedo* (*Clue* in North America) is a murder mystery board game is still popular [99], but represents a type of mystery game in which deduction is more mechanical and can be randomized for each playthrough. Research about murder mystery games focuses on procedural content generation [7, 8, 29, 42, 50, 61, 82] or their educational activities in the classroom [13, 43, 46]. Murder mystery games share similarities with murder or detective-themed *Jubensha* games, yet not all *Jubensha* games are focused on or have murder themes or mysteries, and their storytelling is generally fixed as fiction.

2.1.3 Detectives and Murder in Video Games and Fiction. Video games have used detective settings and metaphors since the early days of CRPGs [52]. Fern á ndez-Vara analyzed early Sherlock Holmes games, finding that the games are adventures that use lock-and-key puzzles to progress the story [33]. In these games, players are not necessarily performing detective work—they are solving designed puzzles—to uncover a mystery [52]. Scholarly work on detective fiction breaks down how mysteries in games work. Todorov introduced the narrative duality of detective novels—the story of the crime and the story of the investigation [87]. Suits argues that the linear structure of detective stories constitutes a game, characterized by one mechanic: the author presenting the mystery with clues in a predetermined sequence, and the reader attempting to solve the mystery before it is revealed [83]. Elements like clues, lists of clues, and maps of the crime scene were included in the detective novels to support the readers in solving the mystery as they reading [48]. This is a critical factor and trend in video games as well. Larsen and Schoenau-Fog found a trend in recent detective games in which the player is given the role of a detective rather than an observer [52]. *Jubensha* scriptwriting is unavoidably influenced by the abundance of forms, styles, and trends in detective novel writing. However, the stories in *Jubensha* games unfold by players reading, performing, and comprehending from various perspectives (i.e., distributed cognition), which adds individual and social aspects to traditional detective novels.

2.2 Game Design and the Design, Dynamics, Experience (DDE) Framework

The DDE Framework [93] is a tool for game design and analysis that formalizes the properties of games. DDE iterates Hunicke's ***Mechanics-Dynamics-Aesthetics* (*MDA*)** Framework [39] to address several shortcomings, including that the MDA Framework was not suited to enacting experience-oriented game design [93]. A principal characteristic of the MDA Framework is how it

[4]In the present research, we provide a ludography for the games considered as part of our dataset. When we cite a game, it is prefixed with a "G."

considers the emotional states of a player (i.e., aesthetics) to be a second-order effect from what the designer can create (i.e., mechanics). In the MDA Framework, *mechanics* are any game element the designer has control over; *dynamics* are emergent game states resulting from how a player makes choices with the mechanics; and the *aesthetics* are the player's resulting emotional state (for which the designer is creating the game).

Corresponding to MDA's mechanics, the DDE Framework identifies DESIGN as anything that the designer has direct control over. This avoids the MDA's name clash— *game mechanics* are more properly defined as designed loops in which players observe the game state, make a decision, enact the decision, and observe the outcome [45, 76]. Under the DDE's DESIGN, there are three subcategories: blueprints, mechanics, and interface. A *blueprint* deals with the game world concept (e.g., terrain, cultures, religions, laws) and style elements (e.g., art, narrative, characters, sound). The [game] *mechanics* describe the implementation game's rules (e.g., object interaction, source architecture) and player choices (e.g., input/output handling). *Interface* describes the concrete elements between the player and game (e.g., graphic assets, sound assets, control responses).

DYNAMICS remain the same between the two—a combination of the designer's intent and the player's choices that lead to particular game states. These are influenced by the DESIGN, but are not directly controlled. The DDE Framework offers three ways of its DYNAMICS—the interactions between the game system and itself, the player with the game system, and the player with the player.

The DDE replaces the MDA Framework's aesthetics with EXPERIENCE—a more direct term for the emotions players feel through play. The DDE organizes EXPERIENCE into three *journeys*: *organoleptic* (i.e., bodily and sensory experiences; e.g., audiovisual), *emotional*, and *intellectual*. Together the three *journeys* are formative for the player's perception of the game.

2.3 Distributed Cognition and Sensemaking

Distributed cognition is a framing for understanding sensemaking activities. *Distributed cognition* conceptualizes the combination of humans and artifacts working in a systemic way over time *as cognition* [38, 40, 73]. Distributed cognition is concerned with how information is gathered, transformed through media, and communicated. A distributed cognition system encompasses people, computation devices, and artifacts: e.g., a cockpit consisting of human pilots, instrument panels, manuals that work together to fly an airplane [40] or a group of people playing a game, who make choices within a set of rules and roles and encode the state of play with their memories and game pieces [73, 76]. These systems are *thinking*—they transform information, move it around, and make decisions. A core part of distributed cognition involves translating and transforming information across media to put it into a form that is most useful. Distributed cognition leads to effective cooperation and can develop a form of sensemaking.

Sensemaking is a process through which an individual or group comes to interpret data about a situation with the intent to take action [96]. Group members collect information from their individual perspectives; transform it through a process of distributed cognition; communicate with each other; collectively develop awareness; and make decisions about how to act [88]. A key part of this process is to translate *information*—raw data that has been fed into a distributed cognition system—into *intelligence*—determining what is true and what matters [88]. Sensemaking processes have been studied in organizations [25] and disaster response [3], as well as multiplayer gaming [88]. Prior work has conceptualized the combination of cooperation and competition in board games as a form of distributed cognition [73] and has highlighted how gathering intelligence and planning is a part of video game design [88].

The present study identifies how *Jubensha* game mechanics are largely a form of group sensemaking through distributed cognition. Players individually acquire information, communicate

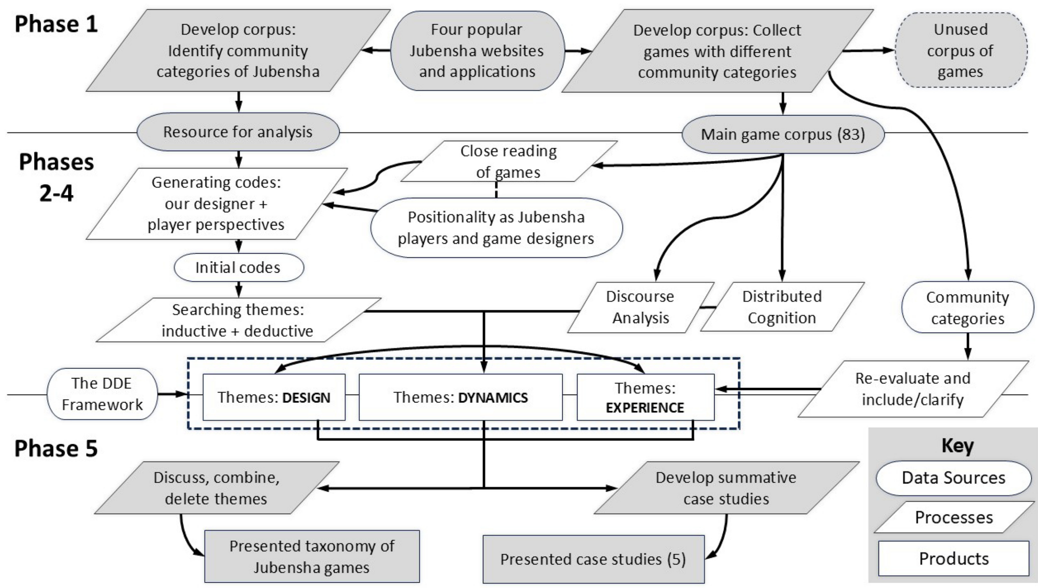

Fig. 2. Diagram of research methodology. **Phase 1:** Data familiarization and development of the game corpus. **Phases 2–4:** Inductive theme development with lenses of close reading of games, discourse analysis, and distributed cognition. **Phase 5:** Inductive and deductive refining, concluding final taxonomy, and developing summative case studies.

about it to develop intelligence, and make decisions. While sensemaking *is a part* of many forms of gameplay, in *Jubensha*, it *is* gameplay.

3 Methodology

We conducted a reflexive thematic analysis [16–18] (e.g., as in Reference [53]) to examine the gameplay, narrative, interfaces, communication within, and features of *Jubensha* games. To understand the data and develop inductive themes, we make use of three lenses, grounded in our positionality: close reading of games [12, 84, 86], discourse analysis [19, 28, 47], and distributed cognition [38, 40]. In this section, we lay out our reflexive thematic analysis process and detail our lenses; Figure 2 diagrams the process and grounds this section. Reflexive thematic analysis consists of six iterative phases, which repeat: (1) familiarizing yourself with the dataset; (2) coding; (3) generating initial themes; (4) developing and reviewing themes; (5) refining, defining, and naming themes; and (6) writing up. We employed inductive coding early on, then used inductive and deductive coding together later. We provide a description of data familiarization (1), a condensed description of inductive theme development (2–4), and our process of inductive/deductive refining (5); the last phase was the process of writing this manuscript. In this report, we also provide an account of the corpus status by describing the many stages it has gone through and its final state.

The present research was performed by five researchers, with the lead author directing the process and engaging most deeply with the material. Two of the team members focused on playing and theme-making *Jubensha* games. Three of the team members focused on discussion and revision of the themes. All team members developed this study and contextualized and developed *Jubensha* game design implications. The project began in May 2023, and analysis was performed until December 2023. In January–February 2024, we wrote the first version of this report. In April–June 2024, we revisited the project and wrote the second version of this report. In September–December

2024, we revised the report and prepared the third revision, which reviewed and provided more detailed taxonomy knowledge products. The work was initially completely inductive, but later came to be a combination of inductive and deductive, described in the following sections.

3.1 Phase 1: Data Familiarization

To build the corpus, we looked at *Jubensha* community websites, gaming platforms, and applications that are considered the most popular [108, 110]:

- 谜圈 (*Miquan*)[5] [78],
- 集石 (*Jishi*)[6] [9],
- 百变大侦探 (*BaiBianDaZhenTan*)[7] [10], and
- 我是谜 (*WoShiMi*)[8] [21].

Our study centered on widely played games on these platforms and websites and we collected *Jubensha* games from these sources to develop the initial game corpus.

Through a combination of reading information from these websites, platforms, and applications as well as playing *Jubensha* games, we include multiple types of data in the corpus. The data include:

- category descriptions and classifications of games from within the community;
- metadata about the games, as pulled from community sites; and
- a number of *Jubensha* themselves (as noted in the Ludography), which, in turn, consist of:
  - game content,
  - scripts, and
  - experience of gameplay.

We have taken the existing experience descriptions, tags, and categories made by community discussions, online apps, and player feedback on platforms into consideration as how they categorize or describe different types of *Jubensha* games. They are used interchangeably, do not always have agreement in the community, do not represent a consistent taxonomy, and are less useful for understanding *Jubensha* from a design perspective. However, because they are widely used, we consider these categories as a resource for building our corpus. The community categories are explained in Table 1.

We intended to create an initial game corpus that includes *Jubensha* games from different existing categories to the greatest extent feasible. Because some *Jubensha* games are designed to be played without a game host while others cannot, we included both kinds. In addition, when developing the game corpus, we found it necessary to investigate physical *Jubensha*. Therefore, both formats were considered equally important in the game corpus and research. Over 100 games were identified in this phase, from which we narrowed our scope.

[5]谜圈, the translated name is *Miquan*, is a *Jubensha* game rating and commenting application developed and published by *Shenzhen Miquan Technology Co., Ltd.*, and more details can be found on their iOS App Store website [78].

[6]集石, the translated name is *Jishi*, is a *Jubensha* game community platform for game rating and social activity, and it has both the MAC application and website, developed and published by *Beijing Jishi Technology Co., Ltd.*; https://www.gstonegames.com; More details can be found on their iOS App Store website [9].

[7]百变大侦探, is developed and published by *Beijing JiuYaoYao Tech CO., LTD.*; Its transliterated name is *BaiBianDaZhenTan* and currently has no English-translated name; https://mszmapp.com. *BaiBianDaZhenTan* is a popular *Jubensha* gaming application that provides a variety of *Jubensha* games that can be purchased and played online with other players, as well as relevant social networking features. It also offers detailed information and databases of playable *Jubensha* games. More details can be found on their iOS App Store website [10].

[8]我是谜, is a *Jubensha* gaming application that developed and published by *Changsha Xiya Technology Co., Ltd.*; Its transliterated name is *WoShiMi* and currently has no English-translated name; https://www.woshimi.cn; More details can be found on their iOS App Store website [21].

Table 1. Community Categories of *Jubensha* Games

CATEGORY	DESCRIPTION
推理: Detective	All games with stories that center solving a mystery.
本格: *Honkaku* / Orthodox	Games deriving from Japanese detective fiction that focus on puzzle solving and realism [14, 70, 85]. ("*Honkaku*" is the Roman character transliteration of the Japanese "ほんかく," and translates to "*orthodox school*" detective fiction.)
变格: *Henkaku* / Unorthodox	Games based on Japanese detective fiction that feature fictional elements that may operate under a logic different from the real world [22, 41, 68, 85]. ("*Henkaku*" is the Roman character transliteration of the Japanese "へんかく": "unorthodox school" detective fiction.)
新本格: *Shin-Honkaku* / New Orthodox	Games that feature fictional elements, but that apply real-world logic. Originates from the Japanese detective fiction genre that focuses on the restoration of the classic rules of Honkaku [32, 66, 68, 70]. (*Shin-Honkaku* is the Roman character transliteration of the Japanese "しんほんかく": "new orthodox school" detective fiction.)
沉浸: Immersive	Games with narratives that the community considers to provide players with strong emotional responses and the perception of being physically present.
架空 / 幻想: Fantasy	Games with aesthetic elements in the narrative that are considered fantasy, such as dragons, castles, knights, etc.
情感: Empathy	Games in which players engage deeply with characters and understand their feelings.
还原: Story-Restoring	Games with stories that center on solving characters' relationships and secrets.
阵营: Competition	Games involve player-versus-player mechanics such as resource acquisition and competition, or team-based mystery-solving competition.
机制: Game Mechanics	Games that use mechanics similar to existing or well-known board games, (e.g., *Werewolf*-style competitions, card deck-building).
现代: Modern	Games that are considered involve modern backgrounds or elements such as cars, aircraft, or computers.
古风 / 民国: Historical Style[a]	Game with fictional stories that are written based on real historical events.
穿越: Time-travel	Games with the narrative that players will experience multiple stories with different era backgrounds during play.
法术: Magic	Games involve magical narrative elements (e.g., spells, wizards, psychics).
日式: Japanese-Style	Games involve narrative elements that are considered Japanese (e.g., *samurai*, *kabuki*).

[a]The Chinese meaning of "古风" refers to the aesthetical setting of ancient China, while "民国" refers to the aesthetical setting of the period from 1911 to 1949 (see: https://projects.iq.harvard.edu/cbdb), which we combined these two categories into one translation as "*Historical Style*" for short.

These categories can overlap and address combinations of setting aesthetics and style of play. English names of the categories are the first author's literal translation, due to the ambiguity of its original Chinese naming of the categories (e.g., translating "还原" into "*[Story]-restoring*").

The initial game corpus was built to include as many games as possible, but we needed to reduce it to be tractable, while remaining a useful sample. We established a template with short descriptions to frame our investigation of *Jubensha* games. We applied the following template to the games in the corpus as selection criteria, to ensure the games in our corpus can encompass as many aspects of each criterion as possible of the following (e.g., games with different numbers of required players or with different existing categories), while reducing games that have overlapping aspects (e.g., games having exactly same existing categories or naming used by the community):

Table 2. Corpus of *Jubensha* Games with Metadata

G	CAT	P	H	#	T
[G1]	*D, Sh, M*		✓	6	7 h
[G2]	*Co, M, G*	✓	✓	8	4 h
[G3]	*E, D, M, He*		✓	6	6 h
[G4]	*E, Hi, D*	✓	✓	6	7 h
[G5]	*D, Sh*			5	Ø
[G6]	*D, Ho*			6	Ø
[G7]	*D, J*			5	Ø
[G8]	*Co, M*	✓	✓	7	5 h
[G9]	*D, M, Co*		✓	6	5 h
[G10]	*D, He, F*		✓	6	8 h
[G11]	*D, F, M*	✓	✓	7	8 h
[G12]	*D, He, M*		✓	6	6 h
[G13]	*D, M, Ho*	✓	✓	7	5 h
[G14]	*D, Sh*	✓	✓	6	5 h
[G15]	*SR, F, J*	✓	✓	6	4 h
[G16]	*Co, Hi*	✓	✓	8	4 h
[G17]	*D, M, SR*		✓	6	3 h
[G18]	*D, M, F*		✓	6	5 h
[G19]	*D, J*		✓	6	7 h
[G20]	*D, Hi, Ma*		✓	7	4 h
[G21]	*D, J*			6	Ø
[G22]	*D, F*	✓	✓	6	5 h
[G23]	*Co, M, D*		✓	7	5 h
[G24]	*D, Hi, Ma*		✓	7	5 h
[G25]	*E, D, I*		✓	5	5 h
[G26]	*D, F, Sh*		✓	6	5 h
[G27]	*E, M, I*		✓	7	4 h
[G28]	*D, T*		✓	5	3 h
[G29]	*D, F*			5	Ø
[G30]	*D, M*		✓	5	5 h
[G31]	*D, F, Ma*	✓	✓	5	Ø
[G32]	*D, Hi, E*	✓	✓	6	5 h
[G33]	*D, SR, Ho*			2	Ø
[G34]	*D, F, Ma*		✓	6	8 h
[G35]	*D, Ho, M*	✓	✓	6	6 h
[G36]	*D, I*	✓	✓	6	4 h
[G37]	*D, Sh*	✓	✓	7	Ø
[G38]	*D, F*		✓	7	5 h
[G39]	*D, F, Ma*	✓	✓	6	4 h
[G40]	*D, F*		✓	7	7 h
[G41]	*D, F, T*			4	Ø
[G42]	*D, M, F*		✓	6	8 h
[G43]	*D, Hi*		✓	6	5 h
[G44]	*D, Hi, Ma*		✓	6	3 h
[G45]	*SR, M*		✓	6	4 h
[G46]	*J, Co, SR*		✓	8	6 h
[G47]	*D, M, He*		✓	6	4 h
[G48]	*D, J*		✓	6	5 h
[G49]	*D, M, Ho*	✓	✓	6	8 h
[G50]	*Co, Hi*		✓	6	4 h
[G51]	*E, J, I*	✓	✓	6	4 h
[G52]	*D, M, He*		✓	6	4 h
[G53]	*D, M, He*		✓	6	6 h
[G54]	*D, Hi, Ma*		✓	6	5 h
[G55]	*D, F, He*			2	Ø
[G56]	*E, M*		✓	6	3 h
[G57]	*Co, Hi, F*		✓	6	5 h
[G58]	*D, M, Ho*		✓	6	6 h
[G59]	*D, M*	✓	✓	7	Ø
[G60]	*D, M*	✓	✓	6	7 h
[G61]	*D, Sh, SR*	✓	✓	6	12 h
[G62]	*D*	✓	✓	6	6 h
[G63]	*D, J, I*		✓	6	5 h
[G64]	*E, M, I*		✓	7	4 h
[G65]	*D, Hi, Ma*		✓	6	4 h
[G66]	*D, M, He*		✓	6	8 h
[G67]	*SR, E, I*		✓	6	5 h
[G68]	*D, F*			5	Ø
[G69]	*D, M, I*		✓	6	4 h
[G70]	*D, F, T*			2	Ø
[G71]	*Co, F, G*	✓	✓	6	3 h
[G72]	*D, M, Ho*	✓	✓	5	5 h
[G73]	*D, M, T*		✓	7	5 h
[G74]	*D, M, E*		✓	6	5 h
[G75]	*D, F, Co*		✓	6	8 h
[G76]	*D, M, Ho*		✓	6	5 h
[G77]	*D, M, G*		✓	6	6 h
[G78]	*D, F*	✓	✓	6	8 h
[G79]	*D, M, E*	✓	✓	7	5 h
[G80]	*D, M, F*		✓	6	5 h
[G81]	*Co, M*		✓	8	6 h
[G82]	*D, SR*	✓	✓	5	6 h
[G83]	*D, E, Ho*			5	Ø

Community category abbreviations (**CAT**): Detective (D), *Honkaku* (Ho), *Henkaku* (He), *Shin-Honkaku* (Sh), Fantasy (F), Empathy (E), Immersive (I), Story Restoring (SR), Competition (Co), Game Mechanics (G), Modern Style (M), Historical Style (Hi), Time-travel (T), Magic (Ma), Japanese Style (J).
G: game identifier. **CAT**: existing community categories (see below). **P**: whether or not the game has a physical version. **H**: whether or not the game uses a game host. #: the number of players. **T**: authors' playtime in hours, rounded **down**, and nil (Ø) for the loss of the time record.

(1) availability (to the researchers);
(2) existing categories used by the publishers or creators;
(3) naming used by the community;
(4) necessity of a game host;
(5) social-material assemblages [100]; and
(6) the number of players.

This template gave us an overview of the corpus and enabled us to filter games for analysis and propose an initial taxonomy. When finalizing the corpus, we supplemented it with games that we did not consider in Phase 1, but that had received media attention or were recommended by other *Jubensha* enthusiasts and players [G8, G31, G36, G37, G46, G49, G61, G64]. The resulting corpus consists of 83 *Jubensha* games, presented in the Ludography and Table 2. This dataset is not exhaustive, but it was sufficiently useful in examining the present phenomena and starting the conversation on the *Jubensha* taxonomy.

3.2 Phases 2–4: Inductive Theme Development

A core part of inductively analyzing our data involved playing the games in the corpus, using three lenses: close reading of games, discourse analysis, and distributed cognition. Close reading enabled us to identify and understand moments of interactivity in games. Discourse analysis was used to analyze materials outside of the games and conversations within play. Distributed cognition was used to identify ways in which information is transformed in game play.

Close reading of games is a process of playing and meticulous documentation, to comprehend the intricacies and nuances inherent in the interactive encounter of a game. Therefore, the process for each game requires a minimum time commitment of 5–6 h. Across 69 games where we kept time records,[9] close-reading time totaled more than 406 h (the playtime of each game included in Table 2). We coded all 83 games and conducted a thorough examination via the close reading of games. As we played the games and reflected on the game experience, initial codes emerged.

We observed the presence of recurring playing strategies and *Jubensha* traits, which we treated as codes. Thus, our codes taken from the player perspective in this phase include the actions that players can take, particular narrative methods or styles encountered, playing patterns emerged, and the observations of other players' communications and conventions, among other things.

At the same time, we aimed to build a taxonomy that would support game design, and so looked inductively for ways in which what we learned might be *useful*. We employed our positionality as game designers to reflect on how and what elements of *Jubensha* play are designed to evoke aesthetic experiences or to motivate particular players' playing strategies, and how that is contextualized within RPGs, board games, and video games. Through this framing, codes that address design strategies for storytelling and to support player interactions, are identified and included.

We used discourse analysis to examine production-side statements and user-side comments, especially how they describe game structure, nomenclature, and playing experience. This served both to understand reception and to understand, translate, and interpret established nomenclature of user-side community and production-side communities. For example, we saw that though the community calls game hosts "*DMs*," their role differs substantially from DMs in other games, thus we decided to use "game host" for greater cross-cultural interpretive accuracy.[10]

Distributed cognition helped in understanding how games unfold among multiple players. Players make decisions about what information to reveal to others, and how. A distributed cognition lens supported considering how game information is represented to players and how they choose to share it with others.

As we developed initial codes, we drew these together in themes that recurred across play experiences. We considered the temporal shape of games and how they supported carefully revealing information.

3.3 Phase 5: Inductive/Deductive Refining

While extracting and categorizing themes from our codes inductively, we aimed to render our discoveries beneficial to developers. Consequently, we sought proven frameworks to incorporate our inductive codes, enhancing the applicability and practicality of our work. We thus transitioned from purely inductive coding to a combination of inductive and deductive coding using established

[9]Data loss of 14 time records occurred and, due to the one-time nature of *Jubensha* games, replaying them would not provide accurate times.

[10]This naming convention of "*DM*" in the *Jubensha* community stems from the Chinese game community's adoption of the naming used in tabletop role-playing games, and was actually borrowed from an external cultural context in English. It is the same word as "*Dungeon Master*," which was originally used in the *Dungeons and Dragons* tabletop role-playing games and refers to the game organizer [23, 97].

game design frameworks. We initially employed the MDA Framework [39], then transitioned its more expressive iteration, the DDE Framework [93], as high-level deductive codes. We considered how our inductive codes fit within the DDE Framework:

— Design—the design element or components of *Jubensha* games that the designer has intentionally constructed;
— Dynamics—the behavior of the *Jubensha* game's different parts interacting with each other and the player while the game is being played; and
— Experience—the perception outcomes players may encounter through *Jubensha* games' dynamics.

Throughout this phase, the researchers actively participated in multiple iterative discussion sessions to investigate the connection between the main themes, the subthemes, our emerging ideas and discussions, and community categories. In reviewing our themes, we considered how community categories can be included that may improve the applicability of our work; Thus, we re-evaluated and clarified their meanings as to how they may be located in our experience themes as subthemes. Meanwhile, we came to understand the value of developing in-depth case studies of exemplars of our themes. As we refined our themes, we created case studies, which appear in Appendix A. Subsequently, codes were integrated into overarching themes, culminating in the production of the report.

4 Analysis and Taxonomy of *Jubensha*

We develop our themes and organize them into design, dynamics, and experience,[11] along with their sub-elements, enabling us to create a taxonomy of *Jubensha*. Figure 3 shows a diagram form of our themes and taxonomy. This section presents our final taxonomy, after multiple rounds of iteration within the phases above.

4.1 Jubensha Design

We incorporated our inductive themes into the DDE framework's design subcategories: blueprint, mechanics, and interface [93]. The gameplay of each *Jubensha* game revolves around a fixed story. The creators of *Jubensha* develop interface surrounding the narrative: e.g., game art, graphic design, audio. The narrative is crafted by *Jubensha* writers and divided into distinct player scripts, each reflecting a unique perspective of the story as one storyline. The game host (if there is one) manuscript lays out the storylines, character relationships, and modus operandi of characters (if applicable). These assets of narrative content, separated scripts, art, and audio, collectively constitute the interface of *Jubensha* game design, encompassing all elements for players to see and hear. Figure 4 shows the interface of a *Jubensha* game [G58] from the application *BaiBianDaZhenTan*.

The design/mechanics of *Jubensha* are constructed around the narrative content and interface; they dissect information from the story and develops the input and output setting, alongside the implementation of game rules for player access. Meanwhile, the interface content transforms and manifests as narrative design, player goals, and other conceptual elements of the *Jubensha* games' blueprint. The story is formalised into event-based phases, although not all games expressly describe them as such; we code these as scenes. The playing process is repeated in each scene, some of which may be timed. During the game, players read their scripts and collect information from each other or from the digital or physical environment (as is applicable), where these allowed ways of interacting with information or game content reveal the *Jubensha* mechanics,

[11] Note that, as with most qualitative work, we use small capitals to denote themes that we found.

***Jubensha* DESIGN**

BLUEPRINT	**MECHANICS**	**INTERFACE**
SCENE(s) **NARRATIVE STRATEGIES Themes:** - LINEAR NARRATIVE - REVERSE - CHRONOLOGY - MULTIPERSPECTIVITY **PLAYER GOALS Themes:** - MULTI-SCENE GOALS - SINGLE-SCENE GOALS - OPTIONAL GOALS	**MECHANICS Themes:** - SCRIPTREADING (读本) - TABLEREADING (小剧场) - INVESTIGATION (搜证) - PLAYER DISCUSSION (讨论) - POLL (投票)	- Story or narrative content - Graphic Assets - Audio Assets - Etc.

Gamehost

***Jubensha* DYNAMICS**

Player ⇔ Game	**Player ⇔ Player**	**Game ⇔ Game**
INDIVIDUAL SENSEMAKING - *Scripted Narrative Involvement* - *Ludic Narrative Involvement* - *Spatial Narrative Involvement*	**COLLABORATIVE SENSEMAKING** - *Affective Narrative Involvement* - *Shared Narrative Involvement* - *Kinaesthetic Narrative Involvement*	**DISTRIBUTED INFORMATION**

***Jubensha* EXPERIENCE**

Emotional Journey	Balanced	*Intellectual Journey*
SOCIAL-FOCUSED Subthemes: - *Romantic* - *Patriotic* - *Sympathetic*	**SOCIAL-FOCUSED** Subthemes: - *Faction-confronting* - *Resource-managing* - *Character-developing*	**DETECTIVE-FOCUSED** Subthemes: - *Honkaku* - *Henkaku* - *Shin-Honkaku*

Fig. 3. *Jubensha* taxonomy and themes from the present research, organized under DESIGN, DYNAMICS, and EXPERIENCE.

which we coded and grouped them into themes. In an ideal playthrough, at the end of the game, players should be able to make sense of the underlying full story, grasp character relationships, resolve potential mysteries, and achieve other goals defined in the game. We coded the PLAYER GOALS of each game as well as the game NARRATIVE STRATEGIES from their use of SCENES' arrangements, and grouped these codes into BLUEPRINT themes. DESIGN themes are directly controlled when making *Jubensha* games, as the themes presented here can be used as the basis for designing *Jubensha* games.

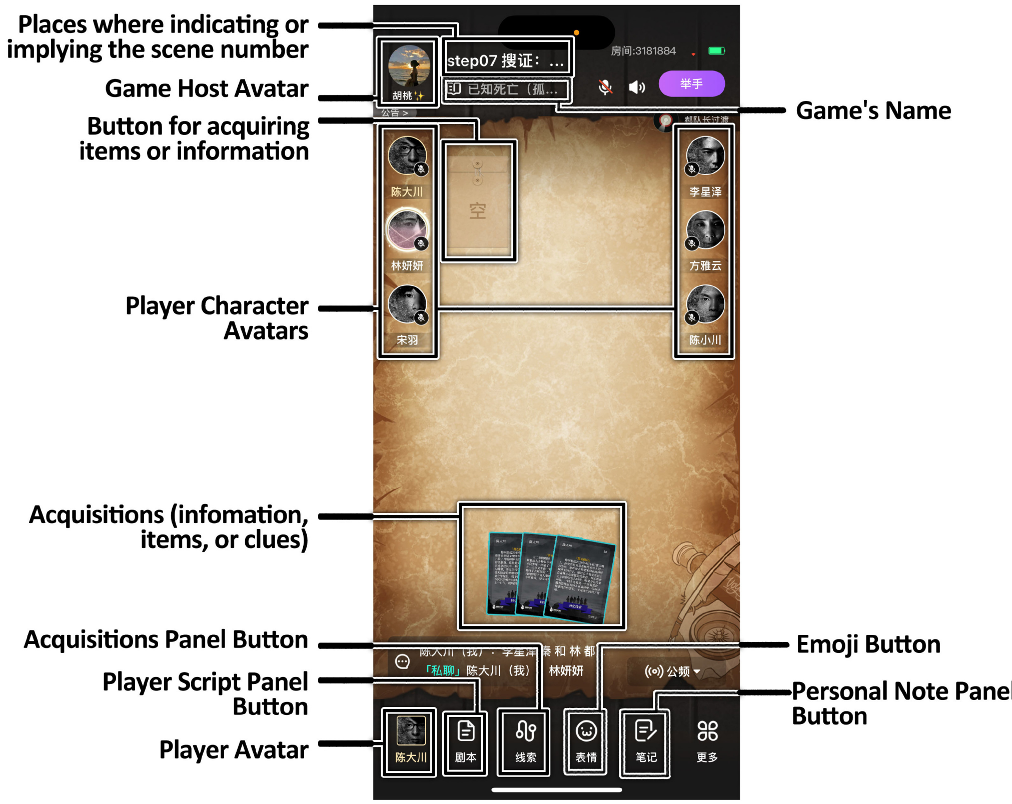

Fig. 4. INTERFACE of the digital version *Jubensha* games, *Death is Debt We All Must Pay* [G58], during player discussion with three acquisitions obtained (located below the center), from the application *BaiBianDaZhen-Tan*. Screenshot taken by the authors.

4.1.1 Mechanics Themes. Each SCENE's contents and interaction procedures vary slightly, yet multiple common MECHANICS can be discerned (Table 3) as creators design the ways players interact with information or game content. The MECHANICS themes are: SCRIPTREADING, TABLEREADING, INVESTIGATION, PLAYER DISCUSSION, and POLL. The number of MECHANICS involved in one SCENE and the time spent on each vary. Figure 1 shows the physical and embodied INTERFACE with integration and arrangement through the MECHANICS of *The Plural Diary of the Yandere Boy* [G37].

4.1.2 Scenes and Narrative Strategies Themes. Each *Jubensha* game normally consists of one or more SCENES. The game's thematic elements are often interconnected across its many SCENES, maintaining consistency until the last SCENE. Players gradually gain information and uncover the story, assembling the story. While the story inside individual SCENES maintains logical coherence, the overall narrative across scenes does not necessarily adhere to a linear structure, reflective of a hypertext structuring of time [56]. Based on the observed games, we describe the use of SCENES as the NARRATIVE STRATEGIES themes of narrative design of the BLUEPRINT in Table 4.

4.1.3 Player Goals Themes. In *Jubensha* games, players are given mandatory and optional goals in their scripts. These goals are designed to help players uncover the story, solve the mystery, or reinforce the player character's characteristics and relationships; however, in games that have

Table 3. *Jubensha* Design Mechanics Themes, Which Reveal Creators' Design of the Ways of Interacting with Information or Game Content, for Players to Access, Perform Sensemaking, and Create Dynamics

Scriptreading (读本)
The players read their scripts individually to ascertain the story from their individual perspectives, an action that is private. These scripts may include either character-specific storylines, which are contingent upon their perspectives, or communal plots in which they collectively partake. While Scriptreading is not timed, the game host's manual may have a recommended or average time for each scene so that the game host can check in on players who read too fast or too slow.
Tablereading (小剧场)
This is a public action—the players read out particular pre-defined lines in their scripts. These scripts may include stage directions such as dialogue, tone, emotions, and facial expressions. The primary function of this mechanics is often enabling players to bring themselves into character and/or immerse themselves in the setting, while certain narrative information can be revealed from this mechanic. Tablereading is not timed.
Investigation (搜证)
The players are timed while they acquire items, information, or clues, either from the environment or through other players within the scene. The rules for how investigation is carried out depend on the game in question—investigation can be private, public, or collaborative, depending on how information gets distributed. In the case of private investigations, these may be carried out through a digital version, where the game supplies the investigating player with information. In digital versions, the acquisitions are represented by images and text. Similarly, a game host may serve the same role for private investigations. In public investigations, all players are privy to whatever information is revealed. In the case of collaborative investigation players are working together—sharing information they have access to or otherwise cooperating to reveal new information. In physical versions, the investigation process may involve interactions with various props such as physical puzzles.
Player Discussion (讨论)
Typically, every scene has one or more sessions that afford players extensive opportunities for discussion. Player discussion may be timed or not timed. There are no explicit guidelines for the discussion. These exchanges often vary and involve negotiations, defenses, and inquiries, where players have the opportunity to share information, deceive, or mislead each other to achieve certain goals.
Poll (投票)
Players vote—either in private or in public to identify the perpetrator or to answer specific questions. Notably, not all scenes have this component, and questions may vary by character.

Table 4. *Jubensha* Narrative Strategies Themes, Which Reveal Creators' Design of Using Scenes for Narrative Design of the Blueprint

Linear Narrative	[G9, G10, G77]
Scenes are sequential. Through scenes, significant events of the tale are revealed, allowing for the gradual unfolding of the characters' experiences.	
Reverse Chronology	[G22, G65, G67]
Scenes are interconnected in a manner that reveals earlier events or characters' experiences in an ordered fashion as the tale unfolds.	
Multiperspectivity	[G17, G40, G48]
Scenes are presented in a disordered manner. In some cases, each scene depicts a distinct occurrence, and player characters might shift while the narrative presents from different perspectives for each player.	

Games that featured the theme are listed next to the theme as examples.

player versus player elements, the goal of some players is to prevent other players from uncovering certain details of the story (e.g., the "murderer" needs to hide key information or mislead other players' discussion and sensemaking). The player typology based on their focus on different out-of-game goals (e.g., playing for fun, achievements, or relaxation) has been explored in a previous study

Table 5. *Jubensha* Player Goals Themes

Multi-Scene Goals
These objectives span multiple scenes and remain active throughout the game. For instance, the character playing the murderer is often instructed to keep direct evidence linked to the crime a secret for the entire game.
Single-Scene Goals
These objectives are relevant only within a specific scene. For example, in scenes involving a backstory set in a fictional world apart from the main game setting, players may be asked to reveal or withhold information within a set timeframe.
Optional Goals
These goals are not mandatory but encourage players to enhance the immersive experience, often through character relationships. For instance, if two characters are romantically involved and one witnesses the other performing a suspicious act (e.g., loitering near cloakrooms, adding powder to someone's meal, entering a restricted area), they are encouraged to keep these actions secret. However, players can choose not to fulfill these goals.

[99]; Therefore, here we focus on player goals revealed from the design, which we recognized and coded into themes of the *Jubensha* blueprint (Table 5).

4.1.4 Role of the Game Host. It is necessary to discuss the game host's role in player interactions. The game host assumes a non-player role, in both digital and offline physical forms of *Jubensha* games. Some games require a game host to lead. Typically, only one host is required. The role of the game host includes managing the playing experience, preventing unexpected actions such as rule deviations from happening, promoting players' discussion, carefully supplying hints, role-playing non-player characters, and assisting players in understanding the puzzles or the overall story at the end. In games with no host, the players collectively undertake the host duties. A game that requires a host usually provides a host manual. The host manual belongs to the narrative assets of interface, which includes scripts of all perspectives, story details, crucial information, and guidance for overseeing the story. The host follows the manual and performs through the mechanics and blueprint of the *Jubensha* game design. Therefore, the role of game host exists in all three subcategories of design, serving as an experienced observer (e.g., as mentioned in Reference [88]) with full knowledge, and an instructor in players' collaborative sensemaking activities.

4.2 Jubensha Dynamics

Jubensha Dynamics address the reasons for and processes of achieving experiences produced by designs through interaction, narrative and sensemaking. The *Jubensha* game design is a structure through which players interact with each other and experience the story. The full plot of each *Jubensha* game's scripts are assigned based on players' choices of characters or perspectives they want to play before starting the game—some games provide character avatars or introductions while some present barely names or numbers to call these perspectives. This section presents our analysis and formed themes of *Jubensha* game's dynamics. Our dynamics include distributed information, individual sensemaking, and collaborative sensemaking.

4.2.1 Distributed Information and Cognition. Distributed information describes the dynamics of the interaction between *Jubensha* game system and itself, produced by the creators' decisions around its design. Each player's script presents different pieces of a complete information picture (e.g., Reference [90]) that must be assembled to understand the storyline. The scripts offer unique perspectives and resemble the standalone narratives of an ensemble cast, where no one player has the complete picture. In addition, some dissected key information is usually placed where it can only be obtained through INVESTIGATION MECHANICS, serving as a public information source. These pieces of information from different scripts or INVESTIGATION are interconnected, forming the DISTRIBUTED INFORMATION DYNAMICS, enacting and creating a distributed cognition frame [38, 40, 73] for players to transform and move information within the group.

Jubensha creators can indirectly control this DYNAMIC through the DESIGN of the total information and its distribution, to ensure the story can be revealed by players' efforts in COLLABORATIVE SENSEMAKING. A Jubensha game whose full story can never be uncovered is usually not considered a good design, which echos the detective fiction's "fair game" notion [49, 52, 91] that the sum of all information provided should always be available for players to uncover and solve the mystery. If the information integrated into the story is hard for INDIVIDUAL SENSEMAKING or COLLABORATIVE SENSEMAKING, then the game's full story may be considered impossible to understand.

4.2.2 Individual Sensemaking and Narrative Involvement. INDIVIDUAL SENSEMAKING describes the DYNAMICS of the interaction between the player and *Jubensha* game system, which involves the individual player's information interpretation and narrative involvement [20]. This DYNAMICS reveals the *Jubensha* game's experiential narrative feature that its narrative is not only scripted but depends upon players' interaction with the game's DESIGN [20, 75], where players' active interpretation and resulting deductions of their sensemaking process further generate the enactment of the game. During INDIVIDUAL SENSEMAKING, three dimensions of narrative involvement occur and promote this DYNAMICS [20]: *Scripted* and *Ludic narrative involvements* from players' interaction with the *Jubensha* story and the DESIGN's rule-system, and *Spatial narrative involvement* from players' interaction with the environment when it's played in physical version or in stores.

DYNAMICS are unpredictable [93], INDIVIDUAL SENSEMAKING is also uncertain and strongly related to the cognitive faculties of the player. Therefore, the challenge for *Jubensha* game creators is to ensure that INDIVIDUAL SENSEMAKING happens effectively through DESIGN. In an ideal running of this DYNAMICS, every player typically has similar proportions. Every player needs to read to perform SCRIPTREADING and INVESTIGATIONS to engage in identifying key information for group communication. This process of extracting and communicating information is a distributed cognition process. Meanwhile, each player character assumes a substantial role in the story, having detailed backgrounds or dedicated scripts, making it uncommon to encounter a *Jubensha* game that explicitly differentiates between main characters and supporting characters.

4.2.3 Collaborative Sensemaking in Comprehending Stories. COLLABORATIVE SENSEMAKING describes the DYNAMICS of the interaction between players, where the information relevant to the general goal of comprehending the story is transforming and spreading among individuals. *Jubensha* engages players in a unique form of COLLABORATIVE SENSEMAKING, where this playing is a process of collecting data collaboratively for group situation awareness to make decisions [3, 25, 88]. Situation awareness is the ability to understand the state of the environment and predict future states [31], whereas, in the *Jubensha* game, it becomes the players' motivation to comprehend the story and predict its unfolding. Whether to uncover the full story or to solve a mystery, players need to take *Jubensha* DESIGN to collect information and participate in concise communication with the group: takeing SCRIPTREADING and INVESTIGATION to understand the situation, taking TABLEREADING and PLAYER DISCUSSION to inform one another of personal status and situation status, and taking POLL as group decisions to respond to the game. In this DYNAMICS, the interactions between players promote *Affective*, *Shared* and *Kinaesthetic narrative involvements* [20], and further influence the narrative quality and *Jubensha* EXPERIENCE.

In addition, as the goal of effective sensemaking is for each team member to have the knowledge and skills to work with others to contribute productively [15, 88], the DESIGN on balancing the distribution of information [73, 90] for different player perspectives in *Jubensha* games is important.

Table 6. *Jubensha* Experience Themes and Subthemes of Aesthetic Elements Describe Ways that Design and Dynamics Can Be Received by Players

Detective-focused Subthemes:	
Honkaku:	[G5–G7, G9, G13, G17, G21, G35, G43, G46, G48, G49, G58, G69, G72, G76, G82]
Henkaku:	[G3, G10–G12, G15, G18, G20, G22, G24, G28, G29, G31, G34, G39, G40, G42, G44, G47, G52–G55, G59, G65, G66, G68, G70, G78, G80]
Shin-Honkaku:	[G1, G14, G19, G25, G26, G30, G33, G36–G38, G41, G60–G62, G73–G75, G77, G79]
Affective-focused Subthemes:	
Romantic:	[G56, G63, G67, G83]
Patriotic:	[G4, G32, G67]
Sympathetic:	[G27, G45, G51, G51, G56, G64]
Social-focused Subthemes:	
Faction-confronting:	[G2, G8, G16, G23, G57, G81]
Resource-managing:	[G2, G8, G71, G81]
Character-developing:	[G16, G50, G71]

Under each theme, examples of games that feature the theme and particular subthemes are presented.

When a *Jubensha* game has one perspective that can gain little information from their scripts to contribute to the collaborative sensemaking, this perspective is a "*marginalized position*" (e.g., Reference [G63]), while the game is considered "unfair." In *Jubensha* games involving players' competition, where the dynamics of player interaction and gameplay become *competitive sensemaking* [11, 57], each party of players should have access to sufficient information to confront others.

4.3 Jubensha Experience

Jubensha experience addresses the range of perception outcomes players may encounter as a result of their design and dynamics. Our resulting experience themes include: detective-focused, affective-focused, and social-focused. These themes encompass their different weighting and primary pursuing of DDE framework experience's *intellectual* (e.g., challenge, discovery, reasoning) and *emotional* (e.g., fear, anger, compassion) *journey* [93]—for its *organoleptic journey* (e.g., audio, visual, smell), we assert all *Jubensha* games may pursuing it more or less. During our analysis, we carefully considered the prior accustomed community categories around aesthetic elements (Table 1). We evaluated some community categories' meanings based on our coding of experience, and include them as subthemes of our themes for improving the applicability of our work (Table 6). Notably, while each *Jubensha* game usually has only one experience theme based on its primary pursuit, the game may have multiplied subthemes of aesthetic elements in different scenes.

4.3.1 Detective-Focused. *Jubensha* games that are detective-focused distinguish themselves through their primary pursuit of the *intellectual journey* [93], as well as the focus on detailed descriptions in scriptreading of crime cases or mysteries. These emphasize design components of investigation and player discussion around information such as deduction regarding modus operandi, puzzles, motivations, and timing. Detective-focused games have an overarching collaborative sensemaking objective, wherein players solve mysteries through poll. A standard arrangement is that each player assumes the role of a detective *and* suspect. Players engaging in a process of reasoning, while also defending against being accused. If a player is designated as the criminal in the game without their knowledge, then they are also encouraged to engage in a process of reasoning to ascertain their culpability without being discovered. The subthemes of DETECTIVE-FOCUSED EXPERIENCE we included are *Honkaku*, *Henkaku*, and *Shin-Honkaku*, where their meanings follow the community categories.

4.3.2 Affective-focused. Jubensha games that are AFFECTIVE-FOCUSED pursue the *emotional journey* [93] and develop a story with substantial interpersonal relationships between characters. Designers take MECHANICS of SCRIPTREADING, TABLEREADING, which emphasize dramatic performance, along with INVESTIGATION to develop a story. In these games, the mysteries are less important and serve as supplementary elements that facilitate the DYNAMICS of player COLLABORATIVE SENSEMAKING of stories. Depending on different affective topics, we define the subthemes as:

— *Romantic*: SCENE(s) that have characters falling in love for players to role-play and experience.
— *Patriotic*: SCENE(s) that have historical stories or fictional events to promote players' reflection and patriotism.
— *Sympathetic*: SCENE(s) that are given to or arise from sympathy, compassion, friendliness, and sensitivity to others' emotions.

4.3.3 Social-focused. Neither pursuing *intellectual* nor *emotional journeys* solely, the pursuit of SOCIAL-FOCUSED *Jubensha* games is usually balanced, while is distinguished by their focus on players' social interaction through the DESIGN MECHANICS and COLLABORATIVE SENSEMAKING DYNAMICS. Few games in the corpus with this theme (e.g., Games [G23, G71]) have mysteries to solve. They invoke INVESTIGATION, DISCUSSION, and POLL extensively to foster player competition, negotiation and satirical or funny stories. INVESTIGATION functions as a way for players to acquire information and resources; PLAYER DISCUSSION is a way to negotiate and trade; and POLL is used to resolve strategic actions or scene-by-scene combat. Depending on the different uses of *Jubensha* DESIGN and DYNAMICS to achieve the SOCIAL-FOCUSED EXPERIENCE, we define the subthemes as:

— *Faction-confronting*: SCENE(s) that have players (actively or passively) choosing their factions and take MECHANICS to achieve their confronting, such as turn-based card battle games, voting battles, or debate competitions.
— *Resource-managing*: SCENE(s) that have particular resources (items, currency, etc.) for players to take MECHANICS to acquire, negotiate, exchange, or auction.
— *Character-developing*: SCENE(s) that provide quests for players to achieve to develop or unlock their character skills that may be used in turn-based battles.

5 Discussion

Jubensha games vary substantially in the characteristics of their story writing, yet they have commonalities in features that can be identified and formalized into themes to build up a taxonomy. With *Jubensha* themes and taxonomy established, we look to answer our research questions in brief here, then expand:

> **RQ1:** What are the key features of *Jubensha* games that enable their definition and analysis?

Through an analysis of 83 *Jubensha* games and their community, we developed a *Jubensha* taxonomy that consists of themes that represent and interpret the key features of these games and are organized into DESIGN, DYNAMICS, and EXPERIENCE.

> **RQ2:** How can the key design features of *Jubensha* games contribute to other games' design?

The DDE Framework serves as a useful tool for designers and researchers. We use our analysis of *Jubensha* games to draw out their particular DESIGN elements—a prewritten story formalized into SCENES, five MECHANICS themes, three NARRATIVE STRATEGIES themes, and three PLAYER GOALS themes—that can be used in service to designing for *Jubensha* games. Our DYNAMIC themes explain how these DESIGN themes enable the interaction and playing of COLLABORATIVE SENSEMAKING in a DISTRIBUTED INFORMATION and cognition environment. Our EXPERIENCE themes offer targets for designers—ways they can frame their own thinking about what they would like for the player to come away with. These taxonomy themes embody *Jubensha* game and playing's flexibility and usability, which allow us to discuss the *Jubensha*'s potential design implications for other games and interdisciplinary explorations.

5.1 Design Implications of Jubensha Games

We aim to support researchers and designers in understanding the *Jubensha* gaming phenomenon and contribute insights into the distinct characteristic implications of *Jubensha* game design. We expect the flexibility and usability of *Jubensha* games' framework may serve a larger interdisciplinary explorations and experience-oriented design purpose, in terms of group experience-making, group sensemaking, and simulator prototyping.

5.1.1 Multiplayer Narrative. One of the main narrative characteristics of *Jubensha* games is the distributed cognition process of exchanging perspectives to piece together the story. The same event presented from one character's perspective can be completely different from other perspectives, and understanding this difference between perspectives can reveal its larger background or story. Taking this design of narrative strategy into multiplayer games can support both the multiplayer differentiability [72], multi-participant interaction of narrative [80], and multiplayer interactive narrative experience design [79], where players experience distinct but connective narratives and can influence the narratives of other players.

5.1.2 Group Experience-making. Plenty of creators use *Jubensha* components and dynamics of group sensemaking to invoke group emotional experiences. The usability of *Jubensha*'s design is reflected in the use of simple paper and pen to write and design scripts can create a quick prototype for a group experience. This unique characteristic of *Jubensha* games has the potential to be used for prototyping various experience-oriented simulators such as group healing games (e.g., Reference [6, 59, 112]), participatory educational narratives (e.g., Reference [109]), or disaster response simulators (e.g., Reference [88, 90]).

5.1.3 Research Potentials in Automatic Game Design (AGD) and Generative Artificial Intelligence (GenAI) for Jubensha Script Generation. The *Jubensha* taxonomy and its COLLABORATIVE SENSEMAKING DYNAMICS we identified are helpful for AGD and GenAI research on creating and evaluating *Jubensha* scripts. AGD is an emerging field in Game AI, which refers to the procedural generation or alternation of an entire game [24]. While many researchers adopt a rule-centric view on games and investigate "*formal abstract design tools*," some scholars are interested in exploring "*non-formalist*" aesthetic-prioritized game design [24]. Our *Jubensha* taxonomy provides a future research agenda for automatic *Jubensha* game design from both formalist and non-formalist perspectives. The DESIGN themes of *Jubensha* can guide "ruleset generations" (i.e., level progress and player objectives in each level). The EXPERIENCE themes, around their different weighting of the *intellectual* or *emotional journey*, can be used as categories for intended exploration, discovery, and sensory experience goals. With the technological advances in natural language processing and retrieval-augmented generation, a future applicable research plan is to utilize the *Jubensha* taxonomy to adapt or convert detective novels into multiplayer, multiround, information-distributed

Jubensha games. Automatic *Jubensha* playtesting could also benefit from the *Jubensha* taxonomy. Previously, Wu et al. presented a preprint on *Large Language Model* behavior and capabilities to serve as a digital detective in multi-agent mystery games [98]. Potential future research could incorporate realistic settings like PLAYER GOALS across SCENES and the COLLABORATIVE SENSEMAKING DYNAMICS among players.

5.2 Dynamics Differences between Physical and Digital Jubensha Games

We studied both physical and digital versions of *Jubensha*. We observe that both show the same outcomes in terms of *Jubensha* game DESIGN and EXPERIENCES. The primary difference between them is the intensity of their DYNAMICS. The MECHANICS and INTERFACE of physical *Jubensha* games usually involve accessories (e.g., physical props or tokens for INVESTIGATION; costumes for TABLEREADING; printed scripts or maps for SCRIPTREADING. These elements strengthen the function of *Jubensha* DESIGN to support the DYNAMICS of player's interactions. Moreover, by seeing players' facial expressions, body language, and performances, the process of sensemaking can be more effective than in digital versions, therefore the DYNAMICS of the playing can be more dramatic or intense. Although in our observations, we did not find the physical versions to have different outcomes in terms of *Jubensha* EXPERIENCE themes, this would be a valuable topic for future examination with user studies.

6 Conclusion

In this article, we analyze the emergent phenomena of *Jubensha* games. We identified themes of *Jubensha* games in terms of their DESIGN, DYNAMICS, and EXPERIENCE, which formalises our taxonomy. Our taxonomy and analysis are grounded in reflexive thematic analysis of 83 *Jubensha* games with lenses from the close reading of games, discourse analysis, and distributed cognition. We offer five summative case studies of *Jubensha* games to exemplify our taxonomy (Appendix A). Further, we discuss design implications for *Jubensha* games for future projects.

Although scholars have applied other game genres to describe *Jubensha* games as their derivative without having agreement in formalizing their gameplay and essential features, the differences and uniqueness of *Jubensha* games are revealed through our analysis and taxonomy. These analyses and the taxonomy of *Jubensha* games aim to provide a descriptive tool and vocabulary for researchers, communities, and designers to facilitate communication and theorize the evolving *Jubensha* gaming phenomenon. Moreover, it also provides the language to formulate *Jubensha* games' particular combination of mechanics and narrative that afford collaborative deciphering, acting, character settings, and comprehending game stories, both elicited and embedded [64, 77].

The study demonstrates that this new gaming phenomenon can be appropriately defined and analyzed in a cross-cultural context. The taxonomy and analysis of *Jubensha* games is also a descriptive tool, vocabulary, or guide for understanding the evolving *Jubensha* communities and their future descendants, facilitating communication when designing or analyzing *Jubensha* games.

Land Acknowledgment (Liang, Chen, Smith, Bohrer)

We recognize the genocide of the Chaubunagungamaug and Hassanamisco Nipmuc Tribe on whose unceded land WPI is located. Because this erasure is not confined to the past, we wish to bring particular awareness to the ongoing efforts of Nipmuc people to regain sovereignty over their ancestral land elsewhere in our state, particularly through the Belchertown-Nipmuc Farm Conservation Alliance https://www.lampsonbrook-landback.org/. The political and financial support of contemporary land-back work is a small step that can be taken to more materially counteract the erasure of the Nipmuc people.

Acknowledgment of Country (Toups Dugas)

I acknowledge the Traditional Owners and Custodians of Country throughout Australia and the lands on which we carry out our research, the Bunurong people of the Kulin Nations. I recognise their continuing connection to the land, waters, and community since time immemorial and that they never ceded sovereignty. I pay my respects to their Elders, past, present, and emerging.

References

[1] Espen Aarseth, Solveig Marie Smedstad, and Lise Sunnanå. 2003. A multidimensional typology of games. In *Proceedings of the DiGRA Conference: Level Up*. Digital Games Research Association, Finland, 48–53.

[2] Sultan A. Alharthi, Olaa Alsaedi, Phoebe O. Toups Dugas, Theresa Jean Tanenbaum, and Jessica Hammer. 2018. Playing to wait: A taxonomy of idle games. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI'18)*. Association for Computing Machinery, New York, NY, 1–15. DOI: https://doi.org/10.1145/3173574.3174195

[3] Sultan A. Alharthi, Nicolas James LaLone, Hitesh Nidhi Sharma, Igor Dolgov, and Phoebe O. Toups Dugas. 2021. An activity theory analysis of search & rescue collective sensemaking and planning practices. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI'21)*. Association for Computing Machinery, New York, NY, Article 146, 20 pages. DOI: https://doi.org/10.1145/3411764.3445272

[4] Sultan A. Alharthi, Phoebe O. Toups Dugas, Olaa Alsaedi, Theresa Jean Tanenbaum, and Jessica Hammer. 2018. *The Pleasure of Playing Less: A Study of Incremental Games through the Lens of Kittens*. Carnegie Mellon University: ETC Press, Pittsburgh, PA. DOI: https://doi.org/10.1184/R1/6686957.v1

[5] Thomas H. Apperley. 2006. Genre and game studies: Toward a critical approach to video game genres. *Simul. Gam.* 37, 1 (2006), 6–23.

[6] Josephine Baird. 2021. Role-playing the self—Trans self-expression, exploration, and embodiment in live action role-playing games. *Int. J. Role-Play.* 11, 11 (Dec. 2021), 94–113. DOI: https://doi.org/10.33063/ijrp.vi11.285

[7] Gabriella Alves Bulhões Barros, Michael Cerny Green, Antonios Liapis, and Julian Togelius. 2019. Who killed albert einstein? From open data to murder mystery games. *IEEE Trans. Games* 11, 1 (Mar. 2019), 79–89. DOI: https://doi.org/10.1109/TG.2018.2806190

[8] Gabriella A. B. Barros, Antonios Liapis, and Julian Togelius. 2016. Murder mystery generation from open data. In *Proceedings of the 7th International Conference on Computational Creativity (ICCC'16)*, Francois Pachet, F. Amilcar Cardoso, Vincent Corruble, and Fiammetta Ghedini (Eds.). Sony CSL, Paris, France, 197–204.

[9] Beijing Jishi Technology Co., Ltd. 2023. 集石. Beijing Jishi Technology Co., Ltd., Beijing, China. Retrieved from https://apps.apple.com/us/app/%E9%9B%86%E7%9F%B3-%E7%BA%BF%E4%B8%8B%E6%B8%B8%E6%88%8F%E7%A4%BE%E4%BA%A4%E5%B9%B3%E5%8F%B0/id1374112517

[10] Beijing JiuYaoYao Tech CO., LTD. 2023. 百变大侦探. Beijing JiuYaoYao Tech CO., LTD., Beijing, China. Retrieved from https://apps.apple.com/us/app/%E7%99%BE%E5%8F%98%E5%A4%A7%E4%BE%A6%E6%8E%A2-%E5%89%A7%E6%9C%AC%E6%9D%80-%E6%88%91%E6%98%AF%E6%8E%A8%E7%90%86%E8%B0%9C%E6%A1%88%E7%8E%8B/id1366306910

[11] Maria Bengtsson and Sören Kock. 2014. Coopetition—Quo vadis? Past accomplishments and future challenges. *Industr. Market. Manage.* 43, 2 (2014), 180–188. DOI: https://doi.org/10.1016/j.indmarman.2014.02.015

[12] Jim Bizzocchi and Theresa Jean Tanenbaum. 2012. Mass effect 2: A case study in the design of game narrative. *Bull. Sci., Technol. Soc.* 32, 5 (2012), 393–404.

[13] Elise A. Blas. 2016. Using a murder mystery to teach evaluation skills: A case study. *Internet Ref. Serv. Quart.* 21, 3-4 (2016), 93–100.

[14] Christopher Bolton, Istvan Csicsery-Ronay, Jr., and Takayuki Tatsumi. 2007. *Robot Ghosts and Wired Dreams: Japanese Science Fiction from Origins to Anime*. University of Minnesota Press, Minneapolis, MN.

[15] Bryan L. Bonner, Michael R. Baumann, and Reeshad S. Dalal. 2002. The effects of member expertise on group decision-making and performance. *Organiz. Behav. Hum. Decision Process.* 88, 2 (2002), 719–736. DOI: https://doi.org/10.1016/S0749-5978(02)00010-9

[16] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitat. Res. Psychol.* 3, 2 (2006), 77–101. DOI: https://doi.org/10.1191/1478088706qp063oa

[17] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitat. Res. Sport, Exerc. Health* 11 (2019), 589–597. Issue 4.

[18] Virginia Braun and Victoria Clarke. 2022. Toward good practice in thematic analysis: Avoiding common problems and be(com)ing a *knowing* researcher. *Int. J. Transgender Health* 24 (2022), 1–6. Issue 1.

[19] Gillian Brown and George Yule. 1983. *Discourse analysis*. Cambridge University Press, Cambridge, UK.

[20] Gordon Calleja. 2013. Narrative involvement in digital games. In *Proceedings of the 12th International Conference on the Foundations of Digital Games (FDG'17)*. Association for Computing Machinery, New York, NY, 9–16. http://www.fdg2013.org/program/papers.html

[21] Changsha Xiya Technology Co., Ltd. 2023. 我是谜. Changsha Xiya Technology Co., Ltd., Changsha, Hunan, China. Retrieved from https://apps.apple.com/us/app/%E6%88%91%E6%98%AF%E8%B0%9C-%E5%89%A7%E6%9C%AC%E6%9D%80-%E7%A0%B4%E6%A1%88%E6%8E%A8%E7%90%86-%E5%BF%AB%E4%B9%90%E4%BA%A4%E5%8F%8B/id1388750151

[22] Nathen Clerici. 2016. Madness, mystery and abnormality in the writing of Yumeno Kyūsaku. *Japan Forum* 28, 4 (2016), 439–464. DOI: https://doi.org/10.1080/09555803.2016.1147484

[23] David "Zeb" Cook. 1989. *Advanced Dungeons & Dragons Dungeon Master Guide.* TSR, Lake Geneva, WI.

[24] Michael Cook and Gillian Smith. 2015. Formalizing non-formalism: Breaking the rules of automated game design. In *Proceedings of the 10th International Conference on the Foundations of Digital Games (FDG'15)*, José P. Zagal, Esther MacCallum-Stewart, and Julian Togelius (Eds.). Society for the Advancement of the Science of Digital Games, Santa Cruz, CA, 5 pages.

[25] Matteo Cristofaro. 2022. Organizational sensemaking: A systematic review and a co-evolutionary model. *Eur. Manage. J.* 40, 3 (2022), 393–405. DOI: https://doi.org/10.1016/j.emj.2021.07.003

[26] Drew Davidson. 2008. Well played: Interpreting *Prince of Persia: The Sands of Time. Games Culture* 3, 3-4 (July 2008), 356–386.

[27] Philippe des Pallières and Hervé Marly. 2001. *The Werewolves of Millers Hollow.* Game [physical].

[28] Jennifer deWinter and Stephanie Vie. 2020. Introduction to Critical Discourse Analysis. Retrieved from http://doi.org/10.17613/f7rq-ah16

[29] Markus Eger. 2020. Murder mysteries: The white whale of narrative generation? *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment.* 210–216. DOI: https://doi.org/10.1609/aiide.v16i1.7432

[30] Christian Elverdam and Espen Aarseth. 2007. Game classification and game design: Construction through critical analysis. *Games Culture* 2, 1 (2007), 3–22.

[31] Mica R. Endsley and Daniel J. Garland. 2000. Theoretical underpinnings of situation awareness: A critical review. In *Situation Awareness Analysis and Measurement* (1st ed.), Mica R. Endsley and Daniel J. Garland (Eds.). 3–21.

[32] FANDOM Books Community of Japanese Mystery Wiki. 2022. "Shinhonkaku Detective Fiction." Retrieved from https://jmystery.fandom.com/wiki/Shinhonkaku_Detective_Fiction

[33] Clara Fernández-Vara. 2013. The game's Afoot: Designing sherlock holmes. In *Proceedings of the DiGRA International Conference: DeFragging Game Studies.* Digital Games Research Association, Finland, 1–14.

[34] Kamiab Ghorbanpour. 2023. Jubensha: The mix of Cluedo, Werewolf and LARP that became a Chinese phenomenon. *Dicebreaker* (Mar. 2023). Retrieved from https://www.dicebreaker.com/categories/roleplaying-game/feature/jubensha-live-action-roleplaying-cluedo-werewolf-chinese-phenomenon

[35] J. Tuomas Harviainen. 2011. Sadomasochist role-playing as live-action role-playing: A trait-descriptive analysis. *Int. J. Role-Play.* 2 (2011), 59–70. DOI: https://doi.org/10.33063/ijrp.vi2.194

[36] J. Tuomas Harviainen, Rafael Bienia, Sarah Lynne Bowman, Simon Brind, Michael Hitchens, Yaraslau I. Kot, Esther MacCallum-Stewart, David W. Simkins, Jaakko Stenros, Ian Sturrock, and Shuo Xiong. 2024. Live-action role-playing games. In *The Routledge Handbook of Role-Playing Game Studies* (1st ed.). Routledge, New York, NY, Chapter 5, 92–114.

[37] William Herkewitz. 2019. How a soviety social experiment became a game for liars. *Popular Mechanics* (28 September 2019). Retrieved from https://www.popularmechanics.com/culture/gaming/a28800275/soviet-mafia-werewolf/

[38] James Hollan, Edwin Hutchins, and David Kirsh. 2000. Distributed cognition: Toward a new foundation for human-computer interaction research. *ACM Trans. Comput.-Hum. Interact.* 7, 2 (June 2000), 174–196. DOI: https://doi.org/10.1145/353485.353487

[39] Robin Hunicke, Marc LeBlanc, and Robert Zubek. 2004. MDA: A formal approach to game design and game research. In *Proceedings of the AAAI Workshop on Challenges in Game AI.* Association for the Advancement of Artificial Intelligence, Washington D.C., 5 pages.

[40] Edwin Hutchins. 1995. How a cockpit remembers its speeds. *Cogn. Sci.* 19, 3 (1995), 265–288. DOI: https://doi.org/10.1016/0364-0213(95)90020-9

[41] Seth Jacobowitz. 2021. The claustrophilic enclosure: Toward a speculative philosophy of perversion in Edogawa Rampo. *Japan Forum* 33, 4 (2021), 756–779. DOI: https://doi.org/10.1080/09555803.2020.1744682

[42] Corinna Jaschek, Tom Beckmann, Jaime A. Garcia, and William L. Raffe. 2019. Mysterious murder - MCTS-driven murder mystery generation. In *Proceedings of the IEEE Conference on Games (CoG'19).* IEEE Press, Piscataway, NJ, 1–8. DOI: https://doi.org/10.1109/CIG.2019.8848000

[43] A. S. Jennings. 2002. Creating an interactive science murder mystery game: The optimal experience of flow. *IEEE Trans. Profess. Commun.* 45, 4 (2002), 297–301. DOI: https://doi.org/10.1109/TPC.2002.805153

[44] Mao Jiayi. 2021. Chinese Gen Z's new obsession: Murder mystery games. *Vogue Business* (14 July 2021). Retrieved from https://www.voguebusiness.com/consumers/chinese-gen-zs-new-obsession-murder-mystery-games

[45] Jesper Juul. 2005. *Half Real: Video Games between Real Rules and Fictional Worlds.* MIT Press, Cambridge, MA.

[46] Rachel Kavanaugh, Zachary Pape, Bonnie LaTourette, and Stefanie Lehmier. 2022. Who killed Mr. Brown? A hospital murder mystery in a pharmacy skills course. *Med. Teacher* 44, 10 (Oct. 2022), 1151–1157. DOI: https://doi.org/10.1080/0142159X.2022.2071690

[47] Reiner Keller. 2011. The sociology of knowledge approach to discourse (SKAD). *Hum. Studies* 34, 1 (2011), 43–65. DOI: https://doi.org/10.1007/s10746-011-9175-z

[48] Stephen Knight. 1980. *Form and Ideology in Crime Fiction*. Palgrave Macmillan, London, UK, Chapter "...done from within"—Agatha Christie's World, 107–134. DOI: https://doi.org/10.1007/978-1-349-05458-9_5

[49] Ronald Knox. 1929. Ten commandments of detective fiction. *Publishers' Weekly* 116 (5 Oct. 1929), 1729. Issue 14.

[50] Max Kreminski, Devi Acharya, Nick Junius, Elisabeth Oliver, Kate Compton, Melanie Dickinson, Cyril Focht, Stacey Mason, Stella Mazeika, and Noah Wardrip-Fruin. 2019. Cozy mystery construction kit: Prototyping toward an AI-assisted collaborative storytelling mystery game. In *Proceedings of the 14th International Conference on the Foundations of Digital Games (FDG'19)*. Association for Computing Machinery, New York, NY, Article 86, 9 pages. DOI: https://doi.org/10.1145/3337722.3341853

[51] Richard W. Kroon. 2010. *A/V A to Z: An Encyclopedic Dictionary of Media, Entertainment and Other Audiovisual Terms*. McFarland, NC.

[52] Bjarke Alexander Larsen and Henrik Schoenau-Fog. 2020. Making the player the detective. In *Proceedings of the 15th International Conference on the Foundations of Digital Games (FDG'20)*. Association for Computing Machinery, New York, NY, 1–8. DOI: https://doi.org/10.1145/3402942.3402969

[53] Shano Liang, Michelle V. Cormier, Phoebe O. Toups Dugas, and Rose Bohrer. 2023. Analyzing trans (Mis)Representation in video games to remediate gender dysphoria triggers. *Proc. ACM Hum.-Comput. Interact.* 7, CHI PLAY, Article 388 (Oct. 2023), 33 pages. DOI: https://doi.org/10.1145/3611034

[54] Ming Liu and Yang Xinhui. 2023. Script Murders: The Form and Design of the Most Popular Tabletop Genre in China. Online Video. Retrievedfromhttps://www.youtube.com/watch?v=LvUoMykLAw4

[55] Yuqiao Liu. 2023. Experiencing China's intangible cultural heritage in role-playing games: Comparative studies between MMORPGs and larps. *Int. J. Role-Play.* 14 (Sep. 2023), 41–46. DOI: https://doi.org/10.33063/ijrp.vi14.358

[56] Marjorie C. Luesebrink. 1998. The moment in hypertext: A brief lexicon of time. In *Proceedings of the 9th ACM Conference on Hypertext and Hypermedia: Links, Objects, Time and Space—Structure in Hypermedia Systems: Links, Objects, Time and Space—Structure in Hypermedia Systems (HYPERTEXT'98)*. Association for Computing Machinery, New York, NY, 106–112. DOI: https://doi.org/10.1145/276627.276639

[57] Eva-Lena Lundgren-Henriksson and Sören Kock. 2016. A sensemaking perspective on coopetition. *Industr. Market. Manage.* 57 (2016), 97–108. DOI: https://doi.org/10.1016/j.indmarman.2016.05.007

[58] Wenhong Luo and Nelson Graburn. 2024. Curating pasts: Musealization and tourism in Chinese cities. *Int. J. Tour. Cities* 10, 2 (May 2024), 509–527. DOI: https://doi.org/10.1108/IJTC-05-2022-0110

[59] Lilis Madyawati and Reza Edwin Sulistyaningtyas. 2020. Local culture games for post-disaster trauma healing in early childhood. In *Proceedings of the 1st Borobudur International Symposium on Humanities, Economics and Social Sciences (BIS-HESS'19)*. Atlantis Press, Dordrecht, The Netherlands, 508–512. DOI: https://doi.org/10.2991/assehr.k.200529.106

[60] Dave McAwesome. ND. The Original Mafia: Dimitry Davidoff's original rules. Retrieved from http://www.maximumawesome.com/articles/mafiaoriginal.htm

[61] Henry Mohr, Markus Eger, and Chris Martens. 2018. Eliminating the impossible: A procedurally generated murder mystery. In *Proceedings of the CEUR Workshop (AIIDE Workshops'18)*. RWTH Aachen University, Aachen, Germany, 7 pages.

[62] Guillaume Montiage. 2001. *Death Wears White*. Game [physical].

[63] Markus Montola. 2008. The invisible rules of role-playing the social framework of role-playing process. *Int. J. Role-Play.* 1 (2008), 22–36. Retrieved from https://journals.uu.se/IJRP/article/view/184

[64] Christopher Moser and Xiaowen Fang. 2015. Narrative structure and player experience in role-playing games. *Int. J. Hum.-Comput. Interact.* 31, 2 (2015), 146–156. DOI: https://doi.org/10.1080/10447318.2014.986639

[65] Florian 'Floyd' Mueller, Martin R. Gibbs, and Frank Vetere. 2008. Taxonomy of exertion games. In *Proceedings of the 20th Australasian Conference on Computer-Human Interaction: Designing for Habitus and Habitat (OZCHI'08)*. Association for Computing Machinery, New York, NY, 263266. DOI: https://doi.org/10.1145/1517744.1517772

[66] NicoNicoPedia. 2021. "新本格". Retrieved from https://dic.nicovideo.jp/a/%E6%96%B0%E6%9C%AC%E6%A0%BC

[67] People Make Games. 2024. The Murder Game Revolution That Has Gripped China. Retrieved from https://www.youtube.com/watch?v=6_dlxbGUNNQ

[68] Baryon Tensor Posadas. 2018. *Double visions, double fictions: The doppelgänger in Japanese film and literature*. University of Minnesota Press, Minneapolis, MN.

[69] Roy Post. 1937. *The Jury Box*. Game [physical].

[70] Ellery Queen. 1997. *Ellery Queen's Japanese Mystery Stories: From Japan's Greatest Detective & Crime Writers*. Tuttle Publishing, North Clarendon, VT.

[71] Gergő Ribáry, László Lajtai, Zsolt Demetrovics, and Aniko Maraz. 2017. Multiplicity: An explorative interview study on personal experiences of people with multiple selves. *Front. Psychol.* 8 (2017), 938.

[72] Mark Riedl, Boyang Li, Hua Ai, and Ashwin Ram. 2011. Robust and authorable multiplayer storytelling experiences. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment.* 189–194.

[73] Melissa J. Rogerson, Martin R. Gibbs, and Wally Smith. 2018. Cooperating to compete: The mutuality of cooperation and competition in boardgame play. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI'18).* Association for Computing Machinery, New York, NY, 1–13. DOI: https://doi.org/10.1145/3173574.3173767

[74] Andrew Rollings and Ernest Adams. 2003. *Andrew Rollings and Ernest Adams on Game Design.* New Riders, Indianapolis, IN.

[75] Marie-Laure Ryan. 2006. *Avatars of Story.* University of Minnesota Press, Minneapolis, MN.

[76] Katie Salen Tekinbaş and Eric Zimmerman. 2003. *Rules of Play: Game Design Fundamentals.* MIT Press, Cambridge, MA.

[77] Edward F. Schneider, Annie Lang, Mija Shin, and Samuel D. Bradley. 2006. Death with a story: How story impacts emotional, motivational, and physiological responses to first-person shooter video games. *Hum. Commun. Res.* 30, 3 (Jan. 2006), 361–375. DOI: https://doi.org/10.1111/j.1468-2958.2004.tb00736.x

[78] Shenzhen Miquan Technology Co., Ltd. 2023. 谜圈. Shenzhen Miquan Technology Co., Ltd., Shenzhen, China. Retrieved from https://apps.apple.com/us/app/%E8%B0%9C%E5%9C%88-%E5%89%A7%E6%9C%AC%E6%9D%80%E7%BB%84%E9%98%9F%E7%A4%BE%E4%BA%A4/id1576501460

[79] Callum Spawforth, Nicholas Gibbins, and David E Millard. 2018. StoryMINE: A system for multiplayer interactive narrative experiences. In *Proceedings of the 11th International Conference on Interactive Digital Storytelling.* Springer International Publishing, Cham, 534–543.

[80] Callum Spawforth and David E. Millard. 2017. A framework for multi-participant narratives based on multiplayer game interactions. In *Proceedings of the 10th International Conference on Interactive Digital Storytelling.* Springer, Cham, Germany, 150–162.

[81] Lizzie Stark. 2012. *Leaving Mundania: Inside the Transformative World of Live Action Role-playing Games.* Chicago Review Press, Chicago, IL.

[82] Andrew Stockdale. 2016. Cluegen: An exploration of procedural storytelling in the format of murder mystery games. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment.* 93–99.

[83] Bernard Suits. 1985. The detective story: A case study of games in literature. *Can. Rev. Compar. Lit./Revue Canadienne de Littérature Comparée* 12, 2 (June 1985), 200–219.

[84] Anne Sullivan and Anastasia Salter. 2017. A taxonomy of narrative-centric board and card games. In *Proceedings of the 12th International Conference on the Foundations of Digital Games* (Hyannis, Massachusetts) *(FDG'17).* Association for Computing Machinery, New York, NY, Article 23, 10 pages. DOI: https://doi.org/10.1145/3102071.3102100

[85] Rebecca Suter. 2011. Science fiction as subversive hypothesis: Henkaku tantei shōsetsu between Entertainment and Enlightenment. *Japan. Studies* 31, 2 (2011), 267–277. DOI: https://doi.org/10.1080/10371397.2011.591775

[86] Theresa Tanenbaum. 2015. Hermeneutic inquiry for digital games research. *Comput. Games J.* 4, 1 (June 2015), 59–80.

[87] Tzvetan Todorov. 2010. The typology of detective fiction (1966). In *Crime and Media: A Reader* (1st ed.), Chris Greer (Ed.). Routledge, London, UK.

[88] Phoebe O. Toups Dugas, William A. Hamilton, and Sultan A. Alharthi. 2016. Playing at planning: Game design patterns from disaster response practice. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY'16).* Association for Computing Machinery, New York, NY, 362–375. DOI: https://doi.org/10.1145/2967934.2968089

[89] Phoebe O. Toups Dugas, Jessica Hammer, William A. Hamilton, Ahmad Jarrah, William Graves, and Oliver Garretson. 2014. A framework for cooperative communication game mechanics from grounded theory. In *Proceedings of the 1st ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play (CHI PLAY'14).* Association for Computing Machinery, New York, NY, 257–266. DOI: https://doi.org/10.1145/2658537.2658681

[90] Phoebe O. Toups Dugas, Andruid Kerne, and William Hamilton. 2009. Game design principles for engaging cooperative play: Core mechanics and interfaces for non-mimetic simulation of fire emergency response. In *Proceedings of the ACM SIGGRAPH Symposium on Video Games (Sandbox'09).* Association for Computing Machinery, New York, NY, 71–78. DOI: https://doi.org/10.1145/1581073.1581085

[91] S. S. Van Dine. 1928. Twenty rules for writing detective stories. *Amer. Mag.* 106 (Sep. 1928), 129–131. Retrieved from https://babel.hathitrust.org/cgi/pt?id=mdp.39015056074373&seq=405

[92] Deborah P. Vossen. 2004. The nature and classification of games. *Avante* 10, 1 (2004), 53–68.

[93] Wolfgang Walk, Daniel Görlich, and Mark Barrett. 2017. Design, dynamics, experience (DDE): An advancement of the MDA framework for game design. In *Game Dynamics: Best Practices in Procedural and Dynamic Game Content Generation*, Oliver Korn and Newton Lee (Eds.). Springer International Publishing, Cham, Germany, 27–45. DOI: https://doi.org/10.1007/978-3-319-53088-8_3

[94] Xinyue Wang and 陆生发. 2024. 剧本杀:作为演述场域的媒介耦合效应生成动力分析. 传媒论坛 7, 5 (2024), 40–45. DOI: https://doi.org/CNKI:SUN:CMLT.0.2024-05-011

[95] Zijun Wang. 2024. "参与式博物馆"理论下的教育活动分析——以"剧本杀"为例. 科学教育与博物馆 10, 2 (2024), 18–24. DOI: https://doi.org/10.16703/j.cnki.31-2111/n.2024.02.004
[96] Karl E. Weick. 1995. *Sensemaking in Organizations.* Foundations for Organizational Science, Vol. 3. Sage, Los Angeles, CA.
[97] William J. White, Nicolas LaLone, and Nicholas J. Mizer. 2022. At the Head of the Table: The TRPG GM as Dramatistic Agent. *RPG学研究 | Japan. J. Analog Role-Play. Game Studies* 3 (Sep. 2022), 46–58. DOI: https://doi.org/10.14989/jarps_3_46
[98] Dekun Wu, Haochen Shi, Zhiyuan Sun, and Bang Liu. 2023. Deciphering Digital Detectives: Understanding LLM Behaviors and Capabilities in Multi-Agent Mystery Games. Retrieved from https://arXiv.2312.00746
[99] Shuo Xiong, Ruoyu Wen, and Huijuan Zheng. 2023. Player Category Research on Murder Mystery Games. *Int. J. Role-Play.* 13 (May 2023), 40–56. DOI: https://doi.org/10.33063/ijrp.vi13.308
[100] José P. Zagal and Sebastian Deterding. 2018. Definitions of "role-playing games." In *Role-playing Game Studies: Transmedia Foundations*, José P. Zagal and Sebastian Deterding (Eds.). Routledge, New York, NY, 19–51.
[101] Li Zhi, Fei Fan, and Wei Jiang. 2023. Enjoyment of Murder Mystery Game Reality Shows: The Influence of Affective Disposition, Suspense and Parasocial Interactions. *Int. J. Commun.* 17, 00 (october 2023), 20.
[102] 何虹雨 and 保虎. 2022. 游戏印记:剧本杀风靡于青年群体的社会学解读. 中国青年研究 1, 9 (2022), 12–17. DOI: https://doi.org/10.19633/j.cnki.11-2579/d.2022.0122
[103] 史可凡 and 赵红勋. 2024. 基于"剧本杀"游戏的社会交往与文化实践研究. 北京文化创意 1 (2024), 47–56. DOI: https://doi.org/CNKI:SUN:BJWH.0.2024-01-007
[104] 季沁园. 2022. 剧本杀:博物馆教育活动新形态. 科学教育与博物馆 8, 2 (2022), 93–99. DOI: https://doi.org/10.16703/j.cnki.31-2111/n.2022.02.013
[105] 张鑫 and 罗勤. 2024. 影游联动, 场域特定与迷影狂欢:作为一种"游戏"的电影剧本杀. 天府新论 3, 2024 (2024), 135–144. DOI: https://doi.org/CNKI:SUN:TFXL.0.2024-03-015
[106] 张雅欣. 2024. 沉浸理论视域下剧本杀行业风险与治理路径析. 新闻研究导刊 15, 5 (2024), 240–242. DOI: https://doi.org/CNKI:SUN:XWDK.0.2024-05-072
[107] 李林 and Wiyi Sun. 2022. 情境与协作:支架式教学理论在历史类博物馆剧本游戏设计中的应用研究. 东南文化 4 (2022), 161–167. DOI: https://doi.org/CNKI:SUN:DNWH.0.2022-04-020
[108] 李韵亭. 2021. 互动演绎下的沉浸式戏剧. Master's thesis. 云南师范大学. DOI: https://doi.org/10.27459/d.cnki.gynfc.2021.000637
[109] 段梦兰, Xiangyue Liu, 陈静, and 张欣迪. 2024. 当健康科普拥抱剧本杀. DOI: https://doi.org/10.28415/n.cnki.njika.2024.000795
[110] 燕道成 and 刘世博. 2021. 青年文化视域下"剧本杀"的兴起与发展趋势. 当代青年研究 6 (2021), 64–70. DOI: https://doi.org/CNKI:SUN:QING.0.2021-06-010
[111] 程晓莉, 詹行, 赵旺来, and 董乔生. 2023. 以休闲之名:"相亲剧本杀"中的青年择偶实践. 当代青年研究 1 (2023), 55–65. DOI: https://doi.org/CNKI:SUN:QING.0.2023-01-004
[112] 罗长青. 2024. 剧本杀作为中华文化国际传播创新路径的学理阐释. *Wuhan University Journal: Philosophy and Social Science* 77, 3 (2024), 20–28. DOI: https://doi.org/10.14086/j.cnki.wujss.2024.03.003
[113] 黄婉婷 and 熊国荣. 2022. 作为界面的身体:线下"剧本杀"游戏中的具身传播及空间建构. 传媒论坛 5, 8 (2022), 4–9. DOI: https://doi.org/CNKI:SUN:CMLT.0.2022-08-002

Ludography

[G1] 严小严; bcc. 2022. 暗蝉. Game [iOS, Android]. Translated name: Cicadas. 神域文化工作室, China.
[G2] DC非凡大陆. 2021. 春运. Game [iOS, Android, physical]. Translated name: Spring Festival Rush. 空然新语, Nanjing, China.
[G3] EmoSpace. 2023. 落沙. Game [iOS, Android]. Translated name: LuoShao. Emo Space, China.
[G4] GoDan发行工作室. 2021. 野火. Game [iOS, Android, physical]. Translated name: Wild Fire. GoDan发行工作室, Shanghai, China.
[G5] Koza. 2020. 棕马馆预告死亡魔术. Game [iOS, Android]. Translated name: Death Notice Magic Show. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G6] Koza. 2020. 看不见的女人. Game [iOS, Android]. Translated name: The Invisible Woman. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G7] Koza. 2021. 长尾の温床. Game [iOS, Android]. Translated name by first author: The Hotbed of Changwei. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G8] PEPPAAAAAAA&GEORGEEEEEE. 2021. 一口鸡腿子. Game [iOS, Android, physical]. Translated name: A Mouthful of Chicken Leg. 这是家大工作室, China.

[G9] 维爷; Star. 2021. 危楼. Game [iOS, Android]. Translated name: Dangerous Building. LARP原创工作室, China.
[G10] Steingate. 2023. 熵增,起源之塔. Game [iOS, Android]. Translated name by first author: Entropy Increase, Tower of Origin. 绿洲发行工作室, China.
[G11] Wangyuchen. 2021. 世界的尽头. Game [iOS, Android, physical]. Translated name: The End of The World. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G12] YC. 2023. 安然街. Game [iOS, Android]. Translated name: AnRan Street. GoDan发行工作室, Shanghai, China.
[G13] yobbo. 2021. 漆昼之瓮. Game [iOS, Android, physical]. Translated name: The Snare of Qizhou City. GoDan发行工作室, Shanghai, China.
[G14] yobbo. 2023. 眠梦不老泉. Game [iOS, Android, physical]. Translated name: Slumber in the Immortal Spring. GoDan发行工作室, Shanghai, China.
[G15] 七庚. 2021. 零子. Game [iOS, Android, physical]. Translated name by first author: Lin Zi. 贱贱与杰西卡工作室, China.
[G16] 不浪大叔. 2022. 大郎该喝药了! Game [iOS, Android, physical]. Translated name: Time for Taking Medicine. 葵花发行工作室, China.
[G17] 两级反转工作室. 2023. 找到我. Game [iOS, Android]. Translated name: Finding Me. 两级反转工作室, China.
[G18] 木橼大叔; 刘奕文. 2022. 波旬. Game [iOS, Android]. Translated name by first author: Boxun. Chengdu LeYun Interactive Network Tech CO., LTD, Chengdu, China.
[G19] 刘律舟. 2023. 立方馆谋杀始末. Game [iOS, Android]. Translated name: Murder In the Cube. 隐匿角色, China.
[G20] 十诫. 2023. 撞灯. Game [iOS, Android]. Translated name: Zhuang Deng. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G21] 千帆过. 2021. 无义即恶. Game [iOS, Android]. Translated name by first author: Wicked of Righteous Deedless. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G22] 千本麻子. 2022. 忒修斯少女. Game [iOS, Android, physical]. Translated name: Theseus Girls. 长歌剧本工作室, China.
[G23] 吴三疯子. 2021. 梼杌. Game [iOS, Android]. Translated name by first author: Tao Wu. 梦起, China.
[G24] 咪咕挑灯. 2023. 风林火山. Game [iOS, Android]. Translated name by first author: Furinkazan. 咪咕挑灯, China.
[G25] 喵桑工作室. 2023. 机械人偶心*2*. Game [iOS, Android]. Translated name: The Soul of Eimy II. GoDan发行工作室, Shanghai, China.
[G26] 塞壬. 2021. *Silence:* 天国. Game [iOS, Android]. Translated name by first author: Silence Heaven. 两级反转工作室, China.
[G27] 白羽非鸦; 姜姜. 2021. 小兔儿乖乖. Game [iOS, Android]. Translated name by first author: Little Bunny Little Kids. 神域文化工作室, China.
[G28] 州一; 安叫兽. 2023. 甜蜜之家. Game [iOS, Android]. Translated name: No Way to Home. 戏精工作室, China.
[G29] 小小子. 2020. *3*月*3*日. Game [iOS, Android]. Translated name by first author: March 3rd. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G30] 尔来. 2021. 杀人回忆. Game [iOS, Android]. Translated name: Murder Memories. GoDan发行工作室, Shanghai, China.
[G31] 王鑫; 峤然. 2021. 年轮. Game [iOS, Android, physical]. Translated name by first author: Annual Rings. 天津剧盟谋杀之谜俱乐部, Tianjin, China.
[G32] 川酱. 2023. 无双. Game [iOS, Android, physical]. Translated name: Wu Shuang. 葵花发行工作室, China.
[G33] 戏子Alice. 2019. 间隙. Game [iOS, Android]. Translated name by first author: Interval. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G34] 我是你的盖世大盖伦儿. 2022. 桃夭, 案台上的女人. Game [iOS, Android]. Translated name by first author: Taoyao, the Woman on Altar. 九云剧制, China.
[G35] 斐然. 2021. 常春藤公寓. Game [iOS, Android, physical]. Translated name: Living. XinMengRenSheng Studio, China.
[G36] 斐然. 2021. 病娇男孩的恋爱日记. Game [iOS, Android, physical]. Translated name: The Love Diary of the Yandere Boy. XinMengRenSheng Studio, Changsha, Hunan, China.
[G37] 斐然. 2021. 病娇男孩的精分日记. Game [iOS, Android, physical]. Translated name: The Plural Diary of the Yandere Boy. XinMengRenSheng Studio, Changsha, Hunan, China.
[G38] 斐然. 2022. 超脱. Game [iOS, Android]. Translated name: Detached. XinMengRenSheng Studio, Changsha, Hunan, China.
[G39] 月莎. 2022. 雾月人鱼潭. Game [iOS, Android, physical]. Translated name: Mermaid. LARP原创工作室, China.
[G40] 楼小葵; 楚棠; 十八; 木马. 2023. 莱茵圣殿的审判夜. Game [iOS, Android]. Translated name: Fallen Angel Judgment Night. GoDan发行工作室, Shanghai, China.
[G41] 朵儿. 2019. 节奏一. Game [iOS, Android]. Translated name: Rhythm One. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.

[G42] 杰尼. 2023. 异想魔术馆. Game [iOS, Android]. Translated name by first author: Fantasy of the Magic House. 葵花发行工作室, China.
[G43] 柒月. 2022. 雪染塘烨村. Game [iOS, Android]. Translated name: Xueran Tangyecun. Shenzhijuzhi Studio, China.
[G44] 梁月. 2023. 苍龙. Game [iOS, Android]. Translated name: Canglong. XinMengRenSheng Studio, Changsha, Hunan, China.
[G45] 楚川. 2023. 月光光奏鸣曲. Game [iOS, Android]. Translated name by first author: YueGuangGuang Sonata. 川衍工作室, China.
[G46] 樱桃. 2021. 艺伎回忆录. Game [iOS, Android]. Translated name by first author: Memoirs of a Geisha. 瓦特工作室, China.
[G47] 汤汤. 2022. 蒙轻纱的处女. Game [iOS, Android]. Translated name: Virgin Veil Pure Sacred. WanYo Space, China.
[G48] 洪公子; 波加曼. 2023. 雨落赤城山. Game [iOS, Android]. Translated name by first author: Rain Falls on Chicheng Hill. 葵花发行工作室, China.
[G49] 洋葱. 2022. 红黑馆事件. Game [iOS, Android, physical]. Translated name: Red and Black Pavilion Incident. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G50] 涂栖. 2023. 苟富贵, 勿相忘. Game [iOS, Android]. Translated name by first author: Don't Forget Your Friends When You Become Rich. XianYan CO., LTD, Nanjing, China.
[G51] 沐黎; 游三. 2023. 美丽新世界. Game [iOS, Android, physical]. Translated name: Brave New World. XianYan CO., LTD, Nanjing, China.
[G52] 漆雕醒. 2022. 红蔷薇杀局. Game [iOS, Android]. Translated name: Red Rose Killing Game. 谜纳剧本社, China.
[G53] 漫小曼. 2023. 山鬼. Game [iOS, Android]. Translated name: The Mountain Spirit. Hubei YueZheQianMing Education Technology Co., Ltd., Hubei, China.
[G54] 狐狸橙子. 2022. *004*鬼娘密室. Game [iOS, Android]. Translated name: Locked Room Mystery of Mother Ghost. 肆言良语, China.
[G55] 王峥. 2019. 晨曦孤儿院. Game [iOS, Android]. Translated name by first author: Chenxi Orphanage. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G56] 甜心. 2021. 假如天堂有假期. Game [iOS, Android]. Translated name: Vacation from Heaven. 海上仙山&可疑的宣发工作室工作室, China.
[G57] 甜梨. 2021. 大势已去. Game [iOS, Android]. Translated name by first author: The tide is over. 十库全书工作室, China.
[G58] 甜甜圈. 2022. 已知死亡. Game [iOS, Android]. Translated name: Death is Debt We All Must Pay. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G59] 甜甜圈. 2023. 档案, 玻璃屋. Game [iOS, Android, physical]. Translated name: The Glass House. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G60] 甜甜圈. 2023. 漫游, 白兔屋. Game [iOS, Android, physical]. Translated name: Kill Alice. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G61] 甜甜圈. 2023. 猫岛谋杀循环. Game [iOS, Android, physical]. Translated name by first author: Murder cycle of Cat Island. 灰烬工作室, China.
[G62] 甜甜圈. 2023. 集会, 魔女屋. Game [iOS, Android, physical]. Translated name: The Witch House. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G63] 蛋挞; 甜甜圈. 2022. 少女の祈祷. Game [iOS, Android]. Translated name: Maiden's Prayer. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G64] 蛋挞; 甜甜圈. 2022. 毛茸茸公寓. Game [iOS, Android]. Translated name by first author: Fluffy House. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G65] 申老师. 2023. 云复往夕. Game [iOS, Android]. Translated name: Clouds Return Every Evening. LARP原创工作室, China.
[G66] 白给. 2023. 本格已死. Game [iOS, Android]. Translated name: No Fundamental Rules. Chengdu Game Ark Culture Communication Co., Ltd., Chengdu, China.
[G67] 挑灯夜未央; 白龙口. 2023. 漠河故事. Game [iOS, Android]. Translated name: The Story of Mohe City. 动物园发行工作室, China.
[G68] 目刮修空调. 2020. *TK*之死. Game [iOS, Android]. Translated name by first author: Death of TK. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G69] 系舟. 2023. 夏日の轻推理. Game [iOS, Android]. Translated name: Something About Summertime. 隐匿角色, China.
[G70] 羲和; 红尘. 2020. 宿命悖论. Game [iOS, Android]. Translated name by first author: Fate Paradox. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.
[G71] 腾腾马. 2022. 尸乐园. Game [iOS, Android, physical]. Translated name: Zombieland. Small Capital Studio, China.

[G72] 花酒间. 2022. 致命协议. Game [iOS, Android, physical]. Translated name: Fatal Agreement. Heimajuzhi Studio and 貔貅文化, China.
[G73] 胡鳕; 落风. 2022. 悬崖边上的$\sqrt{3}$. Game [iOS, Android]. Translated name: Root Three on the Cliff. 葵花发行工作室, China.
[G74] 谭禁. 2023. 漫长的告白. Game [iOS, Android]. Translated name: The Long Confession. 惊蛰推理, China.
[G75] 路人丙. 2023. 女巫请睁眼. Game [iOS, Android]. Translated name: Welcome to the Witch World. 古余工作室, China.
[G76] NIKO; 郭大福. 2020. 追凶手记. Game [iOS, Android]. Translated name: Person of Interest. 白八工作室, China.
[G77] 晚风; 野航. 2023. 海的牙齿神的骨头. Game [iOS, Android]. Translated name: Teeth of the Sea, the Bones of God. 惊蛰推理, China.
[G78] 闻年. 2023. 迷失在时间尽头. Game [iOS, Android, physical]. Translated name: Death & Cycle. 乌鸦坐飞机&剧鱼文创, China.
[G79] 雍久. 2023. 天才在左, 我在右. Game [iOS, Android, physical]. Translated name: Genius is on the Left and I am on the Right. 小生笔笔工作室, China.
[G80] 面包稻草人. 2022. 弗洛伊德之锚. Game [iOS, Android]. Translated name: Fuluoyide Zhi Mao. XinMengRenSheng Studio, Changsha, Hunan, China.
[G81] 魇源. 2022. 火化吧赶紧的. Game [iOS, Android]. Translated name: Cremate it Hurry up. 嗷嗷原创, China.
[G82] 黑SIR. 2022. 树大招风. Game [iOS, Android, physical]. Translated name: Trivisa. 浮生文宝工作室, China.
[G83] 黑猫创意工坊. 2020. 流年. Game [iOS, Android]. Translated name by first author: Years Flow. Beijing JiuYaoYao Tech CO., LTD, Beijing, China.

Appendices

A Case Studies

This section presents five summative case studies from 83 games as exemplars of how *Jubensha* DESIGN bring about DYNAMICS that produce EXPERIENCES. To include as many taxonomy themes as possible in the exemplars as representatives, these selected five *Jubensha* games are, respectively, from different themes or subthemes, and involve:

— One typical *Honkaku detective-focused* game with murder mysteries to solve [G13];
— One *detective-focused* game that shifts its subthemes from *Honkaku* to *Henkaku* during different SCENES [G34];
— One *Shin-Honkaku detective-focused* physical version *Jubensha* game that has no character for player role-playing and encourages sensemaking activities as outsiders [G37];
— One *affective-focused* game with no mysteries to solve [G27]; and
— One *social-focused* game with no murder cases [G8].

A.1 Honkaku Detective-Focused Jubensha Game: The Snare of Qizhou City

***The Snare of Qizhou City* (*SQC*)** [G13] is a DETECTIVE-FOCUSED *Jubensha* game with *Honkaku* subtheme. The narrative of *SQC* revolves around seven player characters who are summoned by an unidentified person to a derelict structure in response to a children's kidnapping incident. The full story of *SQC* is divided into four SCENES arranged in LINEAR NARRATIVE: Discovery of a series of backgrounds and the revelation of each character's secrets; Learning the story about a child bullying incident in a winter camp that happened several weeks ago; Connecting the causes and consequences between the winter camp story and today's kidnapping and finding out the culprit; Attempting to rescue the kidnapped children and failing.

SQC represents *Honkaku* DETECTIVE-FOCUSED *Jubensha* games for its story style and sensemaking direction of solving mysteries. Every player has different game goals about the information they need to hide and facts they need to know from others. In each SCENE of the play, players collect information from their SCRIPTREADING and clue cards acquired from INVESTIGATION, role-playing their characters' quarrels or compromises through TABLEREADING, gradually uncovering the details of the full story through subsequent PLAYER DISCUSSION, and identify the culprits of different occurrences in them through Poll. The background setting and modus operandi of the crime are realistic, where players' mystery-solving relies on their comprehension of principles in the real world.

A.2 Henkaku and Honkaku Detective-Focused Jubensha Game: Taoyao, the Woman on Altar

Taoyao, the Woman on Altar [G34], is a DETECTIVE-FOCUSED *Jubensha* game that consists of three *Honkaku* initial SCENES and one *Henkaku* final SCENE. The game's story follows a REVERSE CHRONOLOGY SCENES-arrangement around six characters who embark on an investigation into a cult situated in an enigmatic village, where they all gradually realize or remember their relationships with events of the cult along with the SCENES' progress.

During the first three scenes, the players acquire information from their SCRIPTREADING and INVESTIGATION about the characters, cult, rumors about missing people, and connections between them. Each player needs to privately reason and confirm whether they are the culprit through their INDIVIDUAL SENSEMAKING of information revealed, meanwhile will act as if they are innocent and conceal unfavorable information from each other throughout TABLEREADING and PLAYER DISCUSSION. Players must answer questions about their characters' roles in the rumors about the cult through POLL. The background and the cases of missing people are all written realistically in the first three SCENES, where players can solve the mystery without any knowledge of the supernatural.

The fourth SCENE revolves around an ongoing murder case and the revelation of several mystical magics and spells. Players discover from their own scripts the spells they previously had employed, respectively. The effects of these spells range from spatial transformation to character resurrection, which is fictionalized and can completely subvert players' previous assumptions and inferences. Once players possess a comprehensive understanding of the mechanics of these spells, they can solve the mystery and identify the only culprit in them through COLLABORATIVE SENSEMAKING.

A.3 Shin-Honkaku Detective-Focused Jubensha Game: The Plural Diary of the Yandere Boy

***The Plural Diary of the Yandere Boy* (*PDYB*)** [G37] is a *Shin-Honkaku* DETECTIVE-FOCUSED *Jubensha* game. Its *Shin-Honkaku* is embodied in how *PDYB* creates a fictionalized explanation of plurality[12] and a pseudo-scientific medicine for plurality, yet the modus operandi of the crimes is realistic.

The story of *PDYB* centers on peculiar criminal events seen through the perspectives of seven alters in a plural system character, in which seven players assume nameless detectors to read alters' diaries. Each alter maintains a diary, which is presented as player scripts, as shown in Figure 5, that can not be shared between players. Given that each of the alters consistently awakens on a designated day of the week, each alter associates their identity with a day of the week, ranging from Monday to Sunday, as their epithets. The game host in this game role-plays and acts as the non-player serial murderer character to bring immersive TABLEREADING experiences. In this setup, the playing throughout *PDYB* revolves around players' engagement in INDIVIDUAL and COLLABORATIVE SENSEMAKING to extract information, exchange perspectives and reveal the story.

The SCENES-arrangement of *PDYB* is MULTIPERSPECTIVITY. The first SCENE commences with the plural system's memories of their abduction, and the objective is for players to determine their

[12]Plurality is the state of having multiple personalities collectively sharing a single body, where a group of personalities is called a system, and each personality is called an alter. It commonly covers the definition of psychological conditions like dissociative identity disorder. For more details, see Reference [71].

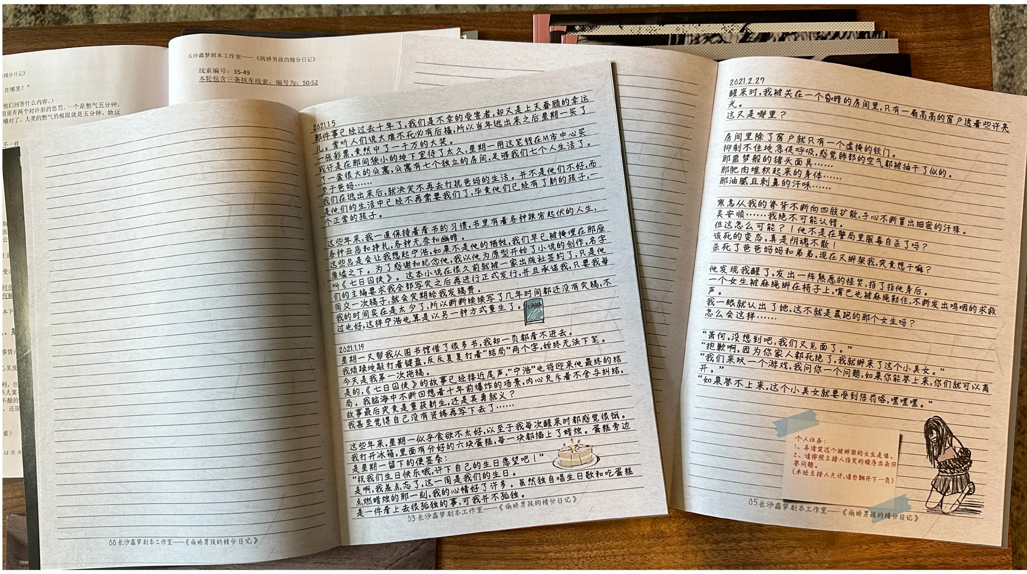

Fig. 5. Photo of the physical player scripts of *The Plural Diary of the Yandere Boy* [G37]. The content of player scripts is presented as different alters' diaries. Photographed by the authors.

perspectives associated with which alter's diary they are given, corresponding with which day of the week. The second SCENE presents a case involving serial murder, while the third SCENE revolves around an escape room scenario related to the plural system. In the fourth SCENE, the script introduces a plot twist that one or more alters of the system have been erased by a fictional medicine in previous scenes, while their diaries are faked by the serial murderer. Players have to reexamine all preceding diaries and reassemble the whole narrative.

A.4 Affective-focused Jubensha Game: Little Bunny Little Kids

Little Bunny Little Kids (LBLK) [G27] is an AFFECTIVE-FOCUSED *Jubensha* game, while its narrative is based on the experiences of seven young patients who are terminally sick with leukemia. The game aims to raise players' *sympathetic* feelings and awareness about the marginalized status of young people with leukemia in the real world.[13] *LBLK* consists of three distinct SCENES in a LINEAR NARRATIVE. The initial SCENE introduces players to a fairy tale setting, where seven players assume the roles of animals; The objective of this scene is to solve a series of riddles collaboratively and employ item cards gained to overcome challenges presented by a wolf.

In the second scene, the background style transitions into a realistic backdrop, where the players become aware that the animals they role-played represent seven youths afflicted with leukemia who are undergoing long-term hospitalization. The scripts assigned to players serve as background stories for each patient, encompassing their encounters. Players can exchange their characters' stories and share their feelings about these characters in PLAYER DISCUSSION. In the concluding scene, players will be informed that their respective characters will progressively confront mortality due to the onset of sickness. Players will be encouraged to embody their

[13] For the same purpose, the publisher and creators of *LBLK* initiated contact with the *China Charities Aid Foundation for Children* and requested a supportive channel through which they can allocate the game's revenues to children who are afflicted with leukemia and need assistance.

characters, articulate their emotions through spontaneous improvisation, and leave comments about the story in TABLEREADING.

In contrast to DETECTIVE-FOCUSED *Jubensha* games, AFFECTIVE-FOCUSED games such as *LBLK* primarily use the DESIGN of *Jubensha* to drive the *affective narrative involvement* [20], conveying the characters' emotions, and evoking emotional resonance from players. For instance, TABLEREADING allows players to experience the roles, while INVESTIGATION enables players to find the memories associated with particular occurrences. In *LBLK*, the narrative content does not involve murder cases or mysteries to solve, which is common to other AFFECTIVE-FOCUSED games, such as the *romantic* story in *Years Flow* [G83] and the *patriotic* story about colonized people in *Wild Fire* [G4].

A.5 Social-focused Jubensha Game: A Mouthful of Chicken Leg

A Mouthful of Chicken Leg (AMCL) [G8] is an SOCIAL-FOCUSED *Jubensha* game centering on player competitiveness, resource management, and puzzle-solving. The story centers on the seven pet characters participating in an animal party and competing for rewards at the owner's house.

The game consists of three distinct SCENES in a LINEAR NARRATIVE. In the first two SCENES, although every character has made trouble in the party, they need to vote to identify "scapegoats" from them. The DESIGN of these SCENES aims to encourage players to accuse each other. All characters except the "scapegoats" will gain cards with random items or in-game currency as rewards. In the final SCENE, players are prompted to organize into two to three groups for a turn-based card game for the victory of *AMCL*. Players can exchange or bid with their gained cards to get powerful item cards before the game starts.

SOCIAL-FOCUSED *Jubensha* games like *AMCL* often utilize *Jubensha* DESIGN to provide diverse interactive rules, gameplay, negotiation, and competition, which are beyond the scope of typical murder mystery solving. In *AMCL*, the process of SCRIPTREADING and TABLEREADING offers side quests, player arguments, and character backgrounds; The INVESTIGATION offers cards that can be used as game items or in-game currency; The PLAYER DISCUSSION mainly facilitates players to engage in negotiation or collaboration; And, the POLL serves the purpose of completing the tasks and gaining rewards.

Received 4 July 2024; revised 17 December 2024; accepted 7 February 2025